

Adjacent-height vanishing for motivic ring spectra

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Abstract

We prove that $K(h)$ -vanishing implies $K(h+1)$ -vanishing for normed motivic spectra over any scheme on which the chosen prime is invertible. The main step is to carry a nilpotence relation in a finite derived quotient through the geometric norm. After extending coefficients, we construct sections of the norms of the Bockstein boundary maps. These show that a coefficient vanishing in the source quotient has norm zero in the target quotient. Applying coefficient projections then decreases the nilpotence exponent. Over separably closed fields, Bott inversion and Hahn's theorem complete the vanishing argument. Dualizability of the finite quotients over their coefficient algebras allows us to descend to arbitrary schemes. We also construct counterexamples at the residue characteristic and explain the $\mathbb{E}_\infty$ counterexamples over $\mathbb{C}$.

1 The question and the coefficient models

Fix a prime p , and write $K^{\text{top}}(h)$ for classical Morava K -theory at p . For an ordinary $\mathbb{E}_\infty$ -ring spectrum R , Hahn's theorem asserts that

$$K^{\text{top}}(h) \wedge R \simeq 0 \implies K^{\text{top}}(h+1) \wedge R \simeq 0.$$

See [11, Theorems 1.1 and 1.3]. Bachmann–Hahn ask for an analogue for normed motivic spectra [6, Question 1.15]. We prove this implication when p is invertible on the base. The counterexamples in the final two sections show that both the normed structure and the restriction on the base matter.

We write $\text{SH}(S)$ for the stable motivic category of a scheme S , and $\text{NAlg}(\text{SH}(S))$ for its category of Sm-normed spectra. For a finite étale map $f : V \rightarrow U$, the motivic norm is a functor

$$f_\otimes : \text{SH}(V) \longrightarrow \text{SH}(U).$$

On a split cover with d sheets it sends $(M_1, \dots, M_d)$ to $M_1 \wedge \dots \wedge M_d$. Write R_U for the pullback of R to U . An Sm-normed spectrum has coherent structure maps $f_\otimes R_V \rightarrow R_U$ for such maps between smooth S -schemes, compatible with composition, base change, and disjoint unions; see [6, §1.4].

An unadorned smash product is taken over the motivic sphere. A tensor product with a subscript is taken over the indicated algebra. All categories and functors are understood in the ∞ -categorical sense. Colimits and tensor products are derived; in a stable category, Map denotes the mapping spectrum. For a motivic spectrum X over S , its ordinary section spectrum is $\Gamma(S, X) = \text{Map}_{\text{SH}(S)}(\mathbf{1}_S, X)$, so $\pi_j \Gamma(S, X) = \pi_{j,0} X$.

Write $L_{(p)} = \mathbb{Z}_{(p)}[t_d \mid d \geq 1]$ for the p -local Lazard ring, with generators chosen so that $t_{p^i-1} = v_i$, as in [6, Proposition 3.7]. The generator t_d has degree d in $L_{(p)}$ and bidegree $(2d, d)$ in MGL. For $n \geq 1$ we use the model

$$K_S(n) = \text{MGL}_{S,(p)}[v_n^{-1}]/(p, t_d \mid d \geq 1, d \neq p^n - 1). \quad (1.1)$$

The quotient means successive cofibers of the specified scalar maps, followed by the filtered colimit over finite lists. Set

$$I_n = (p, v_1, \dots, v_{n-1}), \quad K_S(0) = \text{H}\mathbb{Q}_S.$$

The generators are fixed before base change, so these constructions commute with pullback. We omit the subscript S when the base has been fixed.

Scalar quotients. For a commutative motivic algebra E , an E -module X , and a finite list $I = (b_1, \dots, b_r)$ with $b_i \in \pi_{r_i, s_i} E$, set

$$E/b_i := \text{cofib}(\Sigma^{r_i, s_i} E \xrightarrow{b_i} E), \quad X/I := X \otimes_E (E/b_1) \otimes_E \cdots \otimes_E (E/b_r).$$

The quotient map $X \rightarrow X/I$ is induced by the cofiber maps $E \rightarrow E/b_i$. Thus $\pi_{**}(X/I)$ is the homotopy of a quotient spectrum, whereas $(\pi_{**}X)/I$ is the ordinary quotient of the coefficient module. These groups need not agree.

For a finite list of scalar classes $I = (b_1, \dots, b_r)$, derived completion means

$$X_I^\wedge := \lim_{m \geq 1} X/(b_1^m, \dots, b_r^m),$$

with transition maps induced by the factorizations of successive powers. The notation $X_p^\wedge$ means the case $I = (p)$.

Theorem 1.1. *Let S be a scheme, let p be invertible on S , and let $R \in \text{NAlg}(\text{SH}(S))$. For every finite $h \geq 1$,*

$$K_S(h) \wedge R \simeq 0 \implies (\text{MGL}_{S,(p)}[v_{h+1}^{-1}]/I_{h+1}) \wedge R \simeq 0.$$

In particular,

$$K_S(h) \wedge R \simeq 0 \implies K_S(h+1) \wedge R \simeq 0.$$

The proof over a separably closed field has three stages. Lemma 3.3 first converts $K(h)$ -vanishing into nilpotence of a coefficient in a derived quotient. Proposition 3.4(ii) then allows us to apply the geometric norm to this relation. The coefficient projections of Lemma 2.4 decrease its exponent; iteration proves that the quotient by the Bott class is zero (Proposition 3.8). Finally, Bott inversion shows that the remaining finite quotient is hyperétale local. Its realization is zero by Hahn's theorem, so rigidity implies that the quotient itself is zero.

To pass a vanishing relation through the norm, Proposition 2.1 asks for sections of certain module maps. Lemma 2.2 identifies those maps with relative Thom collapse maps, and Lemma 3.7 constructs their sections. This proves the assertion about the operation in Proposition 3.4(ii).

Section 4 descends the result to arbitrary fields using norms. It then proves vanishing over a scheme by checking the identity of a dualizable module at every residue field. We treat height zero separately before giving the counterexamples.

2 Finite quotients and geometric norms

2.1 Motivic modules over the torsor presheaf

Let E be a normed algebra over a field k , and let C_p be the cyclic group of prime order p . The norm maps for finite étale C_p -torsors commute with pullback. We will use modules over the torsor presheaf to express this compatibility for their coefficients.

For a space-valued presheaf X on Sm_k , define

$$\mathrm{SH}(X) := \lim_{(U \rightarrow X) \in (\mathrm{Sm}_k/X)^{\mathrm{op}}} \mathrm{SH}(U), \quad (2.1)$$

with pullback as the transition functor. The category of elements Sm_k/X has as objects the maps from representable presheaves U to X . Formula (2.1) extends SH from smooth schemes to presheaves by right Kan extension. The resulting category is stable and symmetric monoidal. Its objects are cartesian sections of the fibration classified by $U \mapsto \mathrm{SH}(U)$ [13, Corollary 3.3.3.2].

Write $B_{\mathrm{tors}}C_p$ for the presheaf of finite étale C_p -torsors, and let $\pi : B_{\mathrm{tors}}C_p \rightarrow \mathrm{Spec} k$ be its structure map. An object of $\mathrm{Sm}_k/B_{\mathrm{tors}}C_p$ is a torsor $f : V \rightarrow U$. A morphism to $f' : V' \rightarrow U'$ is a map $a : U \rightarrow U'$ and a C_p -equivariant identification $V \simeq U \times_{U'} V'$. The pullback π^*E is a commutative algebra in $\mathrm{SH}(B_{\mathrm{tors}}C_p)$, with value E_U at $f : V \rightarrow U$. Its module category satisfies

$$\mathrm{Mod}_{\pi^*E}(\mathrm{SH}(B_{\mathrm{tors}}C_p)) \simeq \lim_{(U,f) \in (\mathrm{Sm}_k/B_{\mathrm{tors}}C_p)^{\mathrm{op}}} \mathrm{Mod}_{E_U}(\mathrm{SH}(U)). \quad (2.2)$$

Formation of module objects commutes with limits of symmetric monoidal categories. Hence a module on the left consists of modules M_f over E_U and coherently compatible equivalences $a^*M_{f'} \simeq M_f$.

For $M \in \mathrm{SH}(X)$, write

$$\Gamma(X; M) := \mathrm{Map}_{\mathrm{SH}(X)}(\mathbf{1}_X, M) \simeq \lim_{(U \rightarrow X) \in (\mathrm{Sm}_k/X)^{\mathrm{op}}} \Gamma(U, M_U).$$

For a space-valued presheaf on Sm_k/X , $\Gamma(X; -)$ denotes the limit of the corresponding diagram of spaces.

Let $B_{\mathrm{ét}}C_p$ be the image of $B_{\mathrm{tors}}C_p$ under Nisnevich and $\mathbb{A}^1$ localization in the category of space-valued presheaves on Sm_k . For $M \in \mathrm{SH}(k)$, define the cochain spectrum by the internal Hom

$$M^{B_{\mathrm{ét}}C_p} := \underline{\mathrm{Hom}}_{\mathrm{SH}(k)}(\Sigma_+^\infty B_{\mathrm{ét}}C_p, M).$$

For $T \in \mathrm{SH}(k)$, a map $T \rightarrow M^{B_{\mathrm{ét}}C_p}$ is therefore adjoint to a map $T \wedge \Sigma_+^\infty B_{\mathrm{ét}}C_p \rightarrow M$. The ordinary section spectra satisfy

$$\Gamma(B_{\mathrm{tors}}C_p; \pi^*M) \simeq \Gamma(k, M^{B_{\mathrm{ét}}C_p}). \quad (2.3)$$

To obtain (2.3), express the torsor presheaf as the colimit of its representables and map into $U \mapsto \Gamma(U, M_U)$. Thus modules are defined over the presheaf $B_{\mathrm{tors}}C_p$, whereas their sections in this case can be computed from the motivic space $B_{\mathrm{ét}}C_p$.

For $f : V \rightarrow U$, the relative norm on coefficient modules is the functor

$$\begin{aligned} N_f^E : \mathrm{Mod}_{E_V}(\mathrm{SH}(V)) &\longrightarrow \mathrm{Mod}_{E_U}(\mathrm{SH}(U)), \\ M &\longmapsto E_U \otimes_{f_{\otimes} E_V} f_{\otimes} M. \end{aligned}$$

The norm structure of E defines the algebra map $f_{\otimes} E_V \rightarrow E_U$ used in this tensor product. The functors N_f^E commute with base change by [7, Proposition 7.6(4)]. Consequently, an E -module M over k determines an object $(N_f^E(M_V))_f$ of $\text{Mod}_{\pi^* E}(\text{SH}(B_{\text{tors}} C_p))$. Each N_f^E is symmetric monoidal and preserves zero maps because f has positive degree.

Let A be a normed E -algebra. Represent $x \in \pi_{0,0} A$ by an E -linear map $x : E \rightarrow A$. The maps

$$E_U \simeq N_f^E(E_V) \xrightarrow{N_f^E(x_V)} N_f^E(A_V) \longrightarrow A_U$$

commute with pullback of torsors. Passing to homotopy classes defines the norm operation

$$N_A : \pi_{0,0} A \longrightarrow \pi_{0,0}(A^{B_{\text{ét}} C_p}), \quad (2.4)$$

in the notation of [6, Definition 2.6 and Remark 2.7]. The operation N_A preserves products and the unit.

For the transfer calculation, assume p is invertible in k . The universal stable transfer is

$$\text{tr}_{C_p} : \Sigma_+^{\infty} B_{\text{ét}} C_p \longrightarrow \mathbf{1}_k.$$

At a torsor $f : V \rightarrow U$, the map tr_{C_p} is represented by the composite

$$\Sigma_+^{\infty} U \xrightarrow{\text{tr}_f} \Sigma_+^{\infty} V \longrightarrow \mathbf{1}_k.$$

Here finite étale ambidexterity defines the additive transfer tr_f , and the last arrow is induced by $V \rightarrow \text{Spec } k$. The composites commute with pullback [6, Definition 2.1]. Precomposition defines the coefficient transfer

$$\begin{aligned} \text{Tr}_A : \pi_{0,0} A &\longrightarrow \pi_{0,0}(A^{B_{\text{ét}} C_p}), \\ [x : \mathbf{1}_k \rightarrow A] &\longmapsto [x \circ \text{tr}_{C_p}]. \end{aligned} \quad (2.5)$$

On a split torsor, Tr_A adds the values on the p sheets. In particular, $\text{Tr}_A(1)$ restricts to $p \cdot 1$ at the trivial torsor.

The constant-cochain map

$$\iota_A : \pi_{0,0} A \longrightarrow \pi_{0,0}(A^{B_{\text{ét}} C_p})$$

is induced by $B_{\text{ét}} C_p \rightarrow \text{Spec } k$. The projection formula gives

$$\text{Tr}_A(x) = \iota_A(x) \text{Tr}_A(1).$$

The *transfer ideal*, defined as the ideal generated by the image of Tr_A , is the principal ideal $(\text{Tr}_A(1))$ in $\pi_{0,0}(A^{B_{\text{ét}} C_p})$ [6, Lemma 2.5(4)]. For $x, y \in \pi_{0,0} A$, the norm's additivity formula gives

$$N_A(x + y) - N_A(x) - N_A(y) \in (\text{Tr}_A(1)),$$

so the composite

$$\overline{N}_A : \pi_{0,0} A \longrightarrow \pi_{0,0}(A^{B_{\text{ét}} C_p}) / (\text{Tr}_A(1))$$

is a ring homomorphism [6, Lemma 2.9(4)].

The target of $\overline{N}_A$ is an ordinary quotient of a coefficient ring. To deduce vanishing of $N_A(x)$ in a derived quotient, we will instead apply Proposition 2.1. Its hypothesis is a section in the module category over $B_{\text{tors}} C_p$.

Proposition 2.1. *Let A be a normed E -algebra, and let $c : E \rightarrow C$ be a coaugmented dualizable E -module. Write $C^\vee = \underline{\mathrm{Hom}}_E(C, E)$ for its E -linear dual and $c^\vee : C^\vee \rightarrow E$ for the dual of c . Let*

$$Q \in \mathrm{CAlg}_{\pi^*E}(\mathrm{SH}(B_{\mathrm{tors}}C_p)).$$

Write Q_f for its value at $f : V \rightarrow U$. Suppose the morphism with components

$$N_f^E(C_V^\vee) \otimes_{E_U} Q_f \xrightarrow{N_f^E(c_V^\vee) \otimes 1} Q_f, \quad f : V \rightarrow U,$$

admits a section in $\mathrm{Mod}_Q(\mathrm{SH}(B_{\mathrm{tors}}C_p))$. This requires the Q_f -linear sections to be coherently compatible with pullback. If $x \in \pi_{0,0}A$ has zero image in $\pi_{0,0}(A \otimes_E C)$, its norm has zero image in

$$\pi_0\Gamma(B_{\mathrm{tors}}C_p; \pi^*A \otimes_{\pi^*E} Q).$$

*Here the map from $\pi_{0,0}(A^{B_{\mathrm{et}}C_p})$ to this group is obtained from (2.3) and the morphism $\pi^*A \rightarrow \pi^*A \otimes_{\pi^*E} Q$ induced by the unit of Q .*

Proof. By duality, the image of x in $A \otimes_E C$ corresponds to $C^\vee \xrightarrow{c^\vee} E \xrightarrow{x} A$. The vanishing hypothesis makes this composite null. Pull its nullhomotopy back to V , apply N_f^E , and compose with the normed multiplication of A . The resulting nullhomotopies of

$$N_f^E(C_V^\vee) \xrightarrow{N_f^E(c_V^\vee)} E_U \xrightarrow{N_f^E(x_V)} N_f^E(A_V) \longrightarrow A_U$$

commute with pullback of torsors by norm base change. The last arrow is the norm structure map of the E -algebra A .

Tensor these nullhomotopies with Q_f and compose with the assumed section. The maps $Q_f \rightarrow A_U \otimes_{E_U} Q_f$ are then coherently null. Precomposing with the units of the Q_f proves that the image of $N_A(x)$ in the stated section spectrum is zero. $\square$

For the quotients used below, the maps $c^\vee$ in Proposition 2.1 can be expressed as boundary maps. For a scalar $b : \Sigma^{r,s}E \rightarrow E$, the sequence

$$\Sigma^{r,s}E \xrightarrow{b} E \longrightarrow E/b \xrightarrow{\partial_b} \Sigma^{r+1,s}E$$

defines the *Bockstein boundary* ∂_b . Dualizing this cofiber sequence gives an equivalence

$$(E/b)^\vee \simeq \Sigma^{-r-1,-s}(E/b)$$

under which the dual of $E \rightarrow E/b$ is $\Sigma^{-r-1,-s}\partial_b : \Sigma^{-r-1,-s}(E/b) \rightarrow E$. For a tensor product of scalar cofibers, the dual of the quotient unit is the tensor product over E of these maps. We will split the norms of these boundaries after extending scalars to one algebra in $\mathrm{SH}(B_{\mathrm{tors}}C_p)$. The sections must commute with pullback, as required by Proposition 2.1.

2.2 Relative Thom maps and norms

The Bockstein boundaries above are relative Thom maps. Their norms will be the relative collapse maps for the associated product pairs.

Let $f : V \rightarrow U$ be a C_p -torsor, let $B \hookrightarrow X$ be a finite simplicial pair, and let ℓ be a diagram of E_V -lines on X , with X and B also denoting their realizations. Here a line is an invertible

E_V -module, and the diagram takes values in the space of such modules and their equivalences. Thus $\mathrm{Th}_{E,V}(X, \ell) = \mathrm{colim}_{x \in X} \ell(x)$. The cases needed later are a circle with its vertex and a three-sphere with a contractible hemisphere. Define

$$\mathrm{Th}_{E,V}(X, B; \ell) = \mathrm{cofib}(\mathrm{Th}_{E,V}(B, \ell|_B) \rightarrow \mathrm{Th}_{E,V}(X, \ell)),$$

and let b be its quotient map.

For a nonempty subset $S \subseteq \{1, \dots, p\}$, let X_S be the product with coordinate B on S and coordinate X elsewhere. Set

$$F_B = \bigcup_{S \neq \emptyset} X_S = \bigcup_{j=1}^p X^{j-1} \times B \times X^{p-j}.$$

The group C_p acts on the pair (X^p, F_B) by permuting coordinates.

To interpret the product pair over U , let $\mathcal{O}(C_p)$ be the orbit category, whose objects are C_p/H and whose morphisms are equivariant maps. A genuine C_p -space Y determines the fixed-point functor

$$\mathcal{O}(C_p)^{\mathrm{op}} \longrightarrow \mathcal{S}, \quad C_p/H \longmapsto Y^H.$$

The orbit C_p/H also determines $B_{\mathrm{tors}}H \rightarrow B_{\mathrm{tors}}C_p$. An equivariant map $a : C_p/H \rightarrow C_p/K$ induces, over each torsor $V \rightarrow U$, the map $V \times^{C_p} a$ of associated finite étale schemes. These maps commute with pullback and define the orbit functor on morphisms. Define the associated presheaf over $B_{\mathrm{tors}}C_p$ by

$$\mathrm{Assoc}_{C_p}(Y) := L_{\Sigma} \underset{(C_p/H, x) \in \int_{\mathcal{O}(C_p)} Y(-)}{\mathrm{colim}} (B_{\mathrm{tors}}H \rightarrow B_{\mathrm{tors}}C_p). \quad (2.6)$$

Here the indexing category is the category of elements of the fixed-point functor, and L_{Σ} imposes the condition that disjoint unions of test schemes are sent to products. The colimit is taken in presheaves over $B_{\mathrm{tors}}C_p$ before this localization. In particular,

$$\mathrm{Assoc}_{C_p}(C_p/H) = B_{\mathrm{tors}}H.$$

For a torsor $f : V \rightarrow U$, write

$$\mathrm{Assoc}_{V/U}(Y) := U \times_{B_{\mathrm{tors}}C_p} \mathrm{Assoc}_{C_p}(Y).$$

For a finite C_p -set T this is the finite étale scheme $V \times^{C_p} T$. Its decomposition over the orbits of T is canonical; identifying the component of an orbit with V/H uses a representative whose stabilizer is H .

The norm on coefficient diagrams constructed in Appendix A assigns to ℓ a line diagram

$$\ell_f^{\otimes p} : \mathrm{Assoc}_{V/U}(X^p) \longrightarrow [(T \rightarrow U) \longmapsto \mathrm{Pic}(\mathrm{Mod}_{E_T}(\mathrm{SH}(T)))].$$

This is a map of presheaves on Sm_k/U , where Pic denotes the space of invertible modules and their equivalences. On a finite étale family $(W \rightarrow V, M)$, the construction uses the Weil restriction $Z = R_f W$ and assigns to it the coefficient $N_{f_Z}^E e^* M$. Here $f_Z : Z \times_U V \rightarrow Z$ is projection and $e : Z \times_U V \rightarrow W$ is evaluation. The construction commutes with pullback of torsors. Over the splitting cover V , the line at $(x_g)_{g \in C_p}$ is $\bigotimes_{g \in C_p} g^* \ell(x_g)$.

Write $\mathrm{Th}_{E,U}(X^p, \ell_f^{\otimes p})$ for its Thom colimit, with parameter presheaf $\mathrm{Assoc}_{V/U}(X^p)$. It is computed by applying $q_{\sharp}^E$ to the coefficient on each finite étale orbit scheme $q : W \rightarrow U$, then taking the colimit in $\mathrm{Mod}_{E_U}(\mathrm{SH}(U))$. Here $q_{\sharp}^E$ is left adjoint to pullback of coefficient modules. The same notation applies to invariant subspaces and relative pairs by restriction.

Lemma 2.2 (Relative Thom–norm comparison). *There is a natural equivalence of specified arrows*

$$\begin{array}{ccc} N_f^E \operatorname{Th}_{E,V}(X, \ell) & \xrightarrow{N_f^E(b)} & N_f^E \operatorname{Th}_{E,V}(X, B; \ell) \\ \simeq \downarrow & & \downarrow \simeq \\ \operatorname{Th}_{E,U}(X^p, \ell_f^{\otimes p}) & \longrightarrow & \operatorname{Th}_{E,U}(X^p, F_B; \ell_f^{\otimes p}). \end{array} \quad (2.7)$$

The bottom arrow is the relative Thom collapse. The comparison commutes with pullback of torsors.

Proof. Choose finite simplicial models $B_0 \hookrightarrow X_0$ for the pair and form the simplicial indexing set

$$K_0 = X_0 \amalg_{B_0} (B_0 \star \Delta^0).$$

Extend ℓ to ℓ^0 by assigning zero to the cone vertex and the unique maps to zero on its cone edges. Then

$$\operatorname{colim}_{K_0} \ell^0 \simeq \operatorname{Th}_{E,V}(X, B; \ell),$$

and the inclusion $X_0 \rightarrow K_0$ induces the quotient map b . The indexed-colimit comparison of Lemma A.1 applies to this inclusion, including its noninvertible coefficient maps. It identifies $N_f^E(b)$ with the map of coefficient colimits induced by $X_0^p \rightarrow K_0^p$, formed over the finite étale orbit schemes. The zero-cone product is a coefficient diagram; its transitions need not be equivalences.

Expand K_0^p using $X_0 \leftarrow B_0 \rightarrow B_0 \star \Delta^0$. At a vertex with a cone-vertex coordinate, the coefficient vanishes on its orbit scheme. Indeed, the zero-labelled sheets form an open-and-closed finite étale subcover whose geometric fibres are all nonempty. Decomposing the norm along this subcover and its complement gives a tensor factor equal to the norm of zero along a surjective finite étale map, which is zero. A map from this coefficient colimit to an E_U -module T thus consists of a map from $\operatorname{Th}_{E,U}(X^p, \ell_f^{\otimes p})$ to T and nullhomotopies on its coordinate B -faces. The intersections X_S impose their higher compatibility.

Their homotopy union is F_B as a genuine C_p -space. On underlying spaces this follows because the face inclusions are cofibrations. On C_p -fixed points, the only invariant nonempty coordinate subset is the full set, so the diagram of intersections has fixed-point homotopy colimit $(B^p)^{C_p} = B = F_B^{C_p}$. These are all subgroups of C_p . The mapping spectrum is therefore

$$\operatorname{fib} \left(\operatorname{Map}(\operatorname{Th}_{E,U}(X^p, \ell_f^{\otimes p}), T) \longrightarrow \operatorname{Map}(\operatorname{Th}_{E,U}(F_B, \ell_f^{\otimes p}|_{F_B}), T) \right).$$

This identifies the target with $\operatorname{Th}_{E,U}(X^p, F_B; \ell_f^{\otimes p})$. The inclusion $X_0^p \rightarrow K_0^p$ induces its relative collapse, so the equivalence identifies the specified arrow $N_f^E(b)$ as well. Naturality in Lemma A.1 gives compatibility with pullback of torsors. $\square$

For a line diagram over k and its pullbacks to the torsors, (2.7) gives an equivalence of morphisms in $\operatorname{Mod}_{\pi^*E}(\operatorname{SH}(B_{\operatorname{tors}}C_p))$. The sections of the unit π^*E are $\Gamma(k, E^{B_{\operatorname{ét}}C_p})$, by (2.3).

Lines over the torsor presheaf. To split these arrows, we will trivialize their line diagrams over $B_{\operatorname{tors}}C_p$.

For an ordinary associative ring spectrum T , $\operatorname{GL}_1(T)$ is the union of those components of $\Omega^\infty T$ whose classes are units in $\pi_0 T$; $B\operatorname{GL}_1(T)$ denotes its delooping. Let Q be an associative algebra in $\operatorname{Mod}_{\pi^*E}(\operatorname{SH}(B_{\operatorname{tors}}C_p))$. At a torsor $f : V \rightarrow U$, its value Q_f is an associative E_U -algebra. Define the presheaf of free rank-one right modules by

$$B\operatorname{GL}_1(Q) : (f : V \rightarrow U) \longmapsto B\operatorname{Aut}_{\operatorname{Mod}_{Q_f}(\operatorname{SH}(U))}(Q_f),$$

based at Q_f , with transition maps induced by pullback. Via the category-of-elements equivalence for presheaf slices, we also regard $B\mathrm{GL}_1(Q)$ as a presheaf over $B_{\mathrm{tors}}C_p$: its points over a torsor f are the points of the displayed space. This specifies the target of maps from $\mathrm{Assoc}_{C_p}(X)$ over $B_{\mathrm{tors}}C_p$. Since $B\mathrm{GL}_1(Q)$ takes disjoint unions to products, localization does not change mapping spaces into $B\mathrm{GL}_1(Q)$ or their nullhomotopies. By Yoneda, maps from the free orbit are determined by the value at the trivial torsor; maps from the fixed orbit are determined by sections over $B_{\mathrm{tors}}C_p$.

Limits preserve based loops and units, giving

$$\Omega\Gamma(B_{\mathrm{tors}}C_p; B\mathrm{GL}_1(Q)) \simeq \mathrm{GL}_1(\Gamma(B_{\mathrm{tors}}C_p; Q)). \quad (2.8)$$

For each torsor, the endomorphism spectrum of the right module Q_f is $\Gamma(U, Q_f)$. Taking limits therefore identifies

$$\Gamma(B_{\mathrm{tors}}C_p; Q) \simeq \mathrm{End}_{\mathrm{Mod}_Q(\mathrm{SH}(B_{\mathrm{tors}}C_p))}(Q),$$

where Mod_Q denotes right Q -modules. The endomorphisms here are Q -linear; endomorphisms of the underlying E_U -modules can also include Bockstein operations.

A trivialization of a line diagram restricts to F_B and identifies its relative Thom arrow with the constant-coefficient collapse. Transporting a section of that collapse along (2.7) gives a section of the normed boundary.

2.3 Connective coefficients and finite-cell projections

Postnikov truncation will construct commutative algebra structures on scalar quotients and turn coefficient projections into module maps. For the module maps, we use extension of scalars from ordinary section spectra on the thick subcategory generated by the unit. For an ordinary ring Λ , $H\Lambda$ denotes its Eilenberg–Mac Lane spectrum.

Lemma 2.3 (Connective coefficient quotients). *Let $\mathcal{C}$ be a presentable stable symmetric monoidal category whose tensor product preserves colimits separately, and put $e = \mathrm{End}_{\mathcal{C}}(\mathbf{1}_{\mathcal{C}})$, an ordinary commutative ring spectrum. Suppose that e is connective and*

$$\pi_*e = (\pi_0e)[\beta], \quad |\beta| = 2.$$

Let $b_1, \dots, b_r$ be a regular sequence in π_0e , put $\Lambda = \pi_0e/(\beta, b_1, \dots, b_r)$, and let $g \in \Lambda$. The finite scalar quotients

$$\mathbf{1}_{\mathcal{C}}/(\beta, b_1, \dots, b_r), \quad \mathbf{1}_{\mathcal{C}}/(\beta, b_1, \dots, b_r, g)$$

admit commutative algebra structures with the indicated underlying objects and quotient units. The second quotient uses any chosen lift of g to π_0e ; no regularity of g is required.

Proof. The canonical map $e \rightarrow H\pi_0e$ has a specified nullhomotopy on β , hence factors through an e -module map $e/\beta \rightarrow H\pi_0e$. The polynomial description of π_*e and the cofiber sequence imply that e/β has homotopy only in degree zero. The map induces the identity on π_0e , so it is an equivalence. Regularity of $b_1, \dots, b_r$ similarly identifies

$$e/(\beta, b_1, \dots, b_r) \simeq H\Lambda$$

as e -modules, compatibly with their units.

For the last quotient take the commutative algebra

$$H\Lambda \otimes_{H(\mathbb{Z}[t])} H\mathbb{Z}, \quad t \mapsto g \text{ in } \Lambda, \quad t \mapsto 0 \text{ in } \mathbb{Z}.$$

Here $H(\mathbb{Z}[t])$ denotes the Eilenberg–Mac Lane algebra of the ordinary polynomial ring. Its augmentation module $H\mathbb{Z}$ is the cofiber of multiplication by t on $H(\mathbb{Z}[t])$. Consequently, $H\Lambda \otimes_{H(\mathbb{Z}[t])} H\mathbb{Z}$ has underlying $H\Lambda$ -module $\text{cofib}(g : H\Lambda \rightarrow H\Lambda)$, with the specified quotient unit. This constructs both quotient algebras in $\text{Mod}_e(\text{Sp})$, whether or not g is a zero divisor.

The canonical action of spectra on $\mathcal{C}$ and evaluation $e \otimes \mathbf{1}_{\mathcal{C}} \rightarrow \mathbf{1}_{\mathcal{C}}$ define the colimit-preserving symmetric monoidal functor

$$\text{Mod}_e(\text{Sp}) \longrightarrow \mathcal{C}, \quad M \longmapsto \mathbf{1}_{\mathcal{C}} \otimes_e M.$$

This functor sends the scalar cofibers and their units in $\text{Mod}_e(\text{Sp})$ to the quotients and units in $\mathcal{C}$ stated in the lemma. $\square$

Lemma 2.4 (Finite-cell coefficient projections). *Let E be a commutative motivic algebra over k whose ordinary section spectrum $\Gamma(k, E)$ is connective. Let C, T be E -modules obtained from E by finite sums, cofibers, retracts, and suspensions $\Sigma^{j,0}$, $j \in \mathbb{Z}$. Let Λ be a $\pi_{0,0}E$ -algebra. Suppose that $\Gamma(k, C) \simeq H\Lambda$ as $\Gamma(k, E)$ -modules, where the action on $H\Lambda$ is induced by $\Gamma(k, E) \rightarrow H\pi_{0,0}E \rightarrow H\Lambda$. Suppose also that $\Gamma(k, T)$ is connective and that $\pi_{0,0}T$ is finite free over Λ , with the $\pi_{0,0}E$ -action factoring through Λ . For each basis $e_1, \dots, e_m$ of $\pi_{0,0}T$, there are E -linear maps $T \rightarrow C$ whose maps on $\pi_{0,0}$ send $\sum_i \lambda_i e_i$ to λ_j , for $1 \leq j \leq m$. A map of commutative algebras $E \rightarrow A$ extends each of these maps to $A \otimes_E T \rightarrow A \otimes_E C$.*

Proof. The scalar-extension adjunction is

$$E \otimes_{\Gamma(k, E)} - : \text{Mod}_{\Gamma(k, E)}(\text{Sp}) \rightleftarrows \text{Mod}_E(\text{SH}(k)) : \Gamma(k, -).$$

Its counit

$$E \otimes_{\Gamma(k, E)} \Gamma(k, M) \longrightarrow M$$

is an equivalence for $M = E$. The section functor and extension of scalars are exact. Thus the counit is an equivalence on the thick subcategory generated by E , in particular for $M = C, T$.

Since $\Gamma(k, E)$ is connective, Postnikov truncation is a map of $\Gamma(k, E)$ -modules. Each basis projection on $\pi_{0,0}T$ is $\pi_{0,0}E$ -linear. Applying H and restricting scalars along $\Gamma(k, E) \rightarrow H\pi_{0,0}E$, we obtain maps of $\Gamma(k, E)$ -modules

$$\Gamma(k, T) \longrightarrow H\pi_{0,0}T \longrightarrow H\Lambda.$$

Apply $E \otimes_{\Gamma(k, E)} -$ and use the counit equivalences for T and C . The resulting map $T \rightarrow C$ induces the chosen coefficient projection. The truncation map need not be an equivalence, so $\Gamma(k, T)$ need not be discrete. Tensoring $T \rightarrow C$ over E with A proves the assertion after extension of scalars. $\square$

3 Adjacent-height vanishing over separably closed fields

Let k be a separably closed field of characteristic different from p . All constructions in this section are over k .

Theorem 3.1. *If $h \geq 1$ and $R \in \text{NAlg}(\text{SH}(k))$ satisfy $K(h) \wedge R \simeq 0$, then*

$$(\text{MGL}_{(p)}[v_{h+1}^{-1}]/I_{h+1}) \wedge R \simeq 0. \quad (3.1)$$

In particular, $K(h+1) \wedge R \simeq 0$.

We use periodic cobordism completed at height $h + 1$.

Let PMGL denote periodic algebraic cobordism, with the Thom normed structure of [7, Theorem 16.19]. Its MGL -module decomposition is

$$\text{PMGL} \simeq \bigoplus_{r \in \mathbb{Z}} \Sigma^{2r, r} \text{MGL},$$

with summands indexed by rank. Choose its inverse period $u \in \pi_{-2, -1} \text{PMGL}$ and normalize the height parameters by

$$a_0 = p, \quad a_i = v_i u^{p^i - 1} \ (i \geq 1), \quad J_j = (p, a_1, \dots, a_{j-1}).$$

The a_i have bidegree $(0, 0)$. For a motivic spectrum M , define the completed periodic cobordism spectrum

$$\widehat{\text{PMGL}}_{h+1}(M) := (\text{PMGL}_{(p)}[v_{h+1}^{-1}] \wedge M)_{I_{h+1}}^{\wedge}, \quad \widehat{\text{PMGL}}_{h+1} := \widehat{\text{PMGL}}_{h+1}(\mathbf{1}). \quad (3.2)$$

Completion is taken after smashing with the input M . The unit of R induces the coefficient map

$$\widehat{\text{PMGL}}_{h+1} \longrightarrow \widehat{\text{PMGL}}_{h+1}(R).$$

By [6, Corollary 3.5], both completed spectra inherit normed algebra structures. We can therefore apply the norm operations of Section 2 to these coefficients. For a smooth k -scheme U , a subscript outside parentheses, as in $(\widehat{\text{PMGL}}_{h+1})_U$, denotes pullback of the completed spectrum to U . For every PMGL -module M and $i \geq 1$, multiplication by v_i is the composite

$$\Sigma^{2(p^i - 1), p^i - 1} M \xrightarrow[\simeq]{u^{-(p^i - 1)}} M \xrightarrow{a_i} M.$$

The same factorization for powers identifies the scalar cofibers and completion towers for I_j with those for J_j .

Choose primitive roots $\zeta_{p^r} \in k$ satisfying $\zeta_{p^{r+1}}^p = \zeta_{p^r}$. They specify a generator of $\mathbb{Z}_p(1) := \varprojlim_r \mu_{p^r}(k)$. Under the coefficient comparison

$$\pi_{0, -1}(\text{MGL}_p^{\wedge}) \simeq \pi_{0, -1}(\text{HZ}_p^{\wedge}) \simeq \mathbb{Z}_p(1),$$

the first map is induced by $\text{MGL} \rightarrow \text{HZ}$ and is an isomorphism by [6, Lemma 3.12 and its proof]. The second identifies $\pi_{0, -1}(\text{HZ}_p^{\wedge})$ with $\varprojlim_r H^{0, 1}(k, \mathbb{Z}/p^r) \simeq \varprojlim_r \mu_{p^r}(k)$. Thus the chosen generator lifts to a class in $\pi_{0, -1}(\text{MGL}_p^{\wedge})$. The coefficient map $\text{MGL}_p^{\wedge} \rightarrow \widehat{\text{PMGL}}_{h+1}$ sends the lift to the Bott class $\tau \in \pi_{0, -1} \widehat{\text{PMGL}}_{h+1}$. Set

$$\beta = \tau/u \in \pi_{2, 0} \widehat{\text{PMGL}}_{h+1}.$$

Thus β acts by a scalar map $\Sigma^{2, 0} \widehat{\text{PMGL}}_{h+1} \rightarrow \widehat{\text{PMGL}}_{h+1}$, whose Bockstein boundary has target $\Sigma^{3, 0} \widehat{\text{PMGL}}_{h+1}$. The boundaries for the degree-zero parameters a_i instead have target $\Sigma^{1, 0} \widehat{\text{PMGL}}_{h+1}$.

With this notation, the proof follows the chain

$$\begin{aligned}
& K(h) \wedge R \simeq 0 \\
& \Downarrow \quad \text{Lemma 3.3} \\
& a_h^N = 0 \text{ in } \pi_{0,0}(\widehat{\text{PMGL}}_{h+1}(R)/J_h) \quad \text{for some } N > 0 \\
& \Downarrow \quad \text{reduce modulo } \beta; \quad \text{Proposition 3.8} \\
& \widehat{\text{PMGL}}_{h+1}(R)/(J_h, \beta) \simeq 0 \\
& \Downarrow \quad \begin{array}{l} \text{Bott inversion and realization;} \\ \text{Hahn applied to } K(h) \wedge R = 0 \end{array} \\
& \widehat{\text{PMGL}}_{h+1}(R)/J_{h+1} \simeq 0.
\end{aligned}$$

Proposition 3.4(ii) applies the norm to the nilpotence relation. The coefficient projections of Lemma 2.4 then lower the exponent until $\widehat{\text{PMGL}}_{h+1}(R)/(J_h, \beta)$ vanishes. Section 3.5 proves the last implication. By (3.13), the quotient in Theorem 3.1 is the $r = 0$ summand obtained from the MGL-module decomposition of PMGL after inverting v_{h+1} and quotienting by the homogeneous scalars in I_{h+1} .

3.1 Absolute detection and the initial relation

Lemma 3.2. *For every $j \geq 1$ and $X \in \text{SH}(k)$,*

$$K(j) \wedge X \simeq 0 \iff (\text{MGL}_{(p)}[v_j^{-1}]/I_j) \wedge X \simeq 0.$$

Proof. The motivic Johnson–Wilson module $E(j)$ is obtained from $\text{MGL}_{(p)}[v_j^{-1}]$ by killing the t_d for which $d \notin \{p^i - 1 \mid i \geq 1\}$ and the v_i with $i > j$. We use the resulting $\text{MGL}_{(p)}$ -action. We first show that $E(j)$ and $\text{MGL}_{(p)}[v_j^{-1}]$ have the same acyclic objects. To impose I_j , we then compare the left and right coefficient actions on an absolute smash product and form the relative tensor product.

Detection by the unquotiented theories. The intermediate coefficient algebras remain Landweber exact, and

$$E(j) \otimes_{\text{MGL}_{(p)}} (\text{MGL}_{(p)}/I_j) \simeq K(j).$$

The localization telescope starts an induction identifying the homology of each finite stage with extension of MU_* -coefficients, linearly for its actual scalar action. Each coefficient ring is Landweber exact: $p, v_1, \dots, v_{j-1}$ remain a regular sequence, v_j is a unit, and all other primes are invertible. For the next generator, the short exact coefficient sequence remains exact after tensoring with $\text{MGL}_{**}(Y)$, for every $Y \in \text{SH}(k)$, by the argument of [15, Lemma 5.1]. The long exact sequence of the scalar cofiber therefore identifies its homology with the coefficient cokernel. This identification is MU_* -linear because the cofiber triangle is $\text{MGL}_{(p)}$ -linear. Passing to the filtered colimit gives the Landweber formula for the specified module $E(j)$ and its scalar action [14, Theorem 8.7].

The coefficient extensions to $\mathbb{Z}_{(p)}[v_1, \dots, v_{j-1}, v_j^{\pm 1}]$ and to $\text{MU}_{(p),*}[v_j^{-1}]$ have the same kernel on MU_*MU -comodules [12, Theorem 7.3]. Apply the motivic Landweber formulas in each fixed difference of simplicial degree and twice the weight. They give simultaneous vanishing of the homology groups represented by the two theories.

Apply the same homology comparison to $T^\vee \wedge X$, where T runs through strongly dualizable compact generators. The generators exist after inverting the exponential characteristic [9, Theorem

3.2.1], as p -localization already does. Duality identifies the resulting groups with maps from shifts of T into $E(j) \wedge X$ and $\mathrm{MGL}_{(p)}[v_j^{-1}] \wedge X$. Since the generators detect zero objects,

$$E(j) \wedge X \simeq 0 \iff \mathrm{MGL}_{(p)}[v_j^{-1}] \wedge X \simeq 0. \quad (3.3)$$

The equivalence applies to every $X \in \mathrm{SH}(k)$.

The two actions of the height ideal. Let $\eta : \mathbf{1} \rightarrow \mathrm{MGL}_{(p)}$ be the unit and let

$$\eta_L = 1 \wedge \eta, \quad \eta_R = \eta \wedge 1 : \mathrm{MGL}_{(p)} \longrightarrow \mathrm{MGL}_{(p)} \wedge \mathrm{MGL}_{(p)}$$

be the two coefficient maps. The induced formal group laws F_L, F_R are related by a strict coordinate change $\varphi(T) = T + \sum_{m \geq 1} b_m T^{m+1}$ satisfying

$$\varphi([p]_{F_L}(T)) = [p]_{F_R}(\varphi(T));$$

see [14, Lemma 6.4 and Corollary 6.7]. For $\epsilon = L, R$, our choice of generators gives

$$[p]_{F_\epsilon}(T) \equiv \eta_\epsilon(v_i) T^{p^i} \pmod{\eta_\epsilon(I_i), T^{p^i+1}}$$

by [6, Propositions 3.6(2) and 3.7]. Starting with $\eta_L(p) = \eta_R(p) = p$, compare the coefficient of T^{p^i} after quotienting by the common ideal generated by the preceding parameters. Since $\varphi(T) \equiv T \pmod{T^2}$, the displayed identity gives $\eta_L(v_i) \equiv \eta_R(v_i)$ modulo that ideal. Induction yields

$$(\eta_L(p), \eta_L(v_1), \dots, \eta_L(v_{j-1})) = (\eta_R(p), \eta_R(v_1), \dots, \eta_R(v_{j-1}))$$

in $\pi_{**}(\mathrm{MGL}_{(p)} \wedge \mathrm{MGL}_{(p)})$. Quotienting a module by the scalar maps in either displayed list therefore gives simultaneous vanishing of the two finite Koszul quotients.

More generally, lists of central scalars with the same radical generate the same thick tensor ideal of Koszul objects. A scalar acts square-zero on its own cofiber. If y^N acts by zero on a module, the module is a retract of the cofiber of y^N acting on that module. The cofiber of y^N is a finite extension of cofibers of the action of y . Applying this argument successively to the scalars in each list proves the equality of thick tensor ideals.

Suppose now that $K(j) \wedge X = 0$. On $E(j) \wedge \mathrm{MGL}_{(p)} \wedge X$, the left I_j -quotient is $K(j) \wedge \mathrm{MGL}_{(p)} \wedge X$ and hence vanishes. The right quotient therefore vanishes as well:

$$E(j) \wedge (\mathrm{MGL}_{(p)}/I_j) \wedge X \simeq 0.$$

Applying (3.3) to $(\mathrm{MGL}_{(p)}/I_j) \wedge X$ gives

$$\mathrm{MGL}_{(p)}[v_j^{-1}] \wedge (\mathrm{MGL}_{(p)}/I_j) \wedge X \simeq 0.$$

The relative coefficient quotient. The two-sided bar computing the relative tensor product has, after smashing with X , degree- q term

$$\mathrm{MGL}_{(p)}[v_j^{-1}] \wedge \mathrm{MGL}_{(p)}^{\wedge q} \wedge (\mathrm{MGL}_{(p)}/I_j) \wedge X, \quad q \geq 0.$$

The face maps use the two $\mathrm{MGL}_{(p)}$ -module actions and multiplication in $\mathrm{MGL}_{(p)}$; the degeneracy maps insert its unit. Each term is the zero absolute smash product just obtained, further smashed with $\mathrm{MGL}_{(p)}^{\wedge q}$. Realizing the bar therefore gives

$$(\mathrm{MGL}_{(p)}[v_j^{-1}] \otimes_{\mathrm{MGL}_{(p)}} (\mathrm{MGL}_{(p)}/I_j)) \wedge X \simeq (\mathrm{MGL}_{(p)}[v_j^{-1}]/I_j) \wedge X \simeq 0.$$

This proves the forward implication. The reverse implication follows by taking the additional cofibers in (1.1). $\square$

Lemma 3.3. *Under the hypothesis of Theorem 3.1, there is an $N > 0$ such that the coefficient a_h^N has zero image in $\pi_{0,0}(\widehat{\text{PMGL}}_{h+1}(R)/J_h)$.*

Proof. Since $\widehat{\text{PMGL}}_{h+1}(R)$ is an R -module, the hypothesis of Theorem 3.1 implies $K(h) \wedge \widehat{\text{PMGL}}_{h+1}(R) = 0$. Lemma 3.2 makes $(\text{MGL}_{(p)}[v_h^{-1}]/I_h) \wedge \widehat{\text{PMGL}}_{h+1}(R)$ zero. Applying the two-sided bar construction as in that lemma gives

$$(\text{MGL}_{(p)}[v_h^{-1}]/I_h) \otimes_{\text{MGL}_{(p)}} \widehat{\text{PMGL}}_{h+1}(R) \simeq (\widehat{\text{PMGL}}_{h+1}(R)/J_h)[a_h^{-1}] \simeq 0.$$

The composite of the quotient unit with the localization telescope is null. Since the motivic sphere is compact over a field, the unit becomes null at a finite stage of the telescope. Thus for some $N > 0$,

$$a_h^N = 0 \quad \text{in } \pi_{0,0}(\widehat{\text{PMGL}}_{h+1}(R)/J_h). \quad (3.4)$$

Equation (3.4) refers to the image of the coefficient a_h^N under the quotient unit. The argument does not require a multiplication on $\widehat{\text{PMGL}}_{h+1}(R)/J_h$. $\square$

3.2 The operation on a derived quotient

For a line bundle L on a motivic space Y , the orientation gives $c_1(L) \in \pi_{-2,-1}(\widehat{\text{PMGL}}_{h+1}^Y)$. Define the normalized Euler class by

$$z(L) := u^{-1}c_1(L) \in \pi_{0,0}(\widehat{\text{PMGL}}_{h+1}^Y).$$

Here cochains on Y are defined by the same internal function spectrum as in Section 2. Let F be the formal group law over $\pi_{0,0}\widehat{\text{PMGL}}_{h+1}$ characterized by $z(L \otimes L') = F(z(L), z(L'))$. The chosen primitive root ζ_p gives a faithful character $C_p \rightarrow \mathbb{G}_m$. Write z for the normalized Euler class of the associated universal line bundle on $B_{\text{ét}}C_p$. For $m \geq 0$, the series $[m]_F(Z)$ is the m -fold sum of Z under F , and $X +_F Y$ means $F(X, Y)$. The field k is p -good: it has p -cohomological dimension zero and contains all p -power roots of unity. Thus the coefficient and completed homology calculations of [6, Lemma 3.12 and §4] apply. They give the cochain presentation

$$\pi_{0,0}(\widehat{\text{PMGL}}_{h+1}^{B_{\text{ét}}C_p}) = (\pi_{0,0}\widehat{\text{PMGL}}_{h+1})[[z]]/([p]_F(z)). \quad (3.5)$$

As a $\widehat{\text{PMGL}}_{h+1}$ -module, $\widehat{\text{PMGL}}_{h+1}^{B_{\text{ét}}C_p}$ is free on $1, z, \dots, z^{p^{h+1}-1}$, of rank p^{h+1} . The same calculations show that the weight-zero homotopy of $\widehat{\text{PMGL}}_{h+1}$ and $\widehat{\text{PMGL}}_{h+1}^{B_{\text{ét}}C_p}$ is connective and concentrated in even degrees, with coefficient rings $(\pi_{0,0}\widehat{\text{PMGL}}_{h+1})[\beta]$ and $\pi_{0,0}(\widehat{\text{PMGL}}_{h+1}^{B_{\text{ét}}C_p})[\beta]$.

Lemma 2.3 equips $\widehat{\text{PMGL}}_{h+1}/(J_h, \beta)$ with a commutative coefficient algebra structure and its prescribed quotient unit. Modulo J_h , the p -series is divisible by Z^{p^h} . Choose a series

$$g_h(Z) \in (\pi_{0,0}\widehat{\text{PMGL}}_{h+1})[[Z]], \quad Z^{p^h}g_h(Z) \equiv [p]_F(Z) \pmod{J_h}.$$

The reduction of $g_h(Z)$ modulo J_h is unique and has constant coefficient a_h . Substitution in (3.5) gives $g_h(z) \in \pi_{0,0}(\widehat{\text{PMGL}}_{h+1}^{B_{\text{ét}}C_p})$. By (2.3), this class determines the $\pi^*\widehat{\text{PMGL}}_{h+1}$ -linear endomorphism

$$g_h : \pi^*\widehat{\text{PMGL}}_{h+1} \longrightarrow \pi^*\widehat{\text{PMGL}}_{h+1} \quad \text{in } \text{Mod}_{\pi^*\widehat{\text{PMGL}}_{h+1}}(\text{SH}(B_{\text{tors}}C_p)).$$

At $f : V \rightarrow U$, its value is multiplication by the pullback $g_h(f) \in \pi_{0,0}(\widehat{\text{PMGL}}_{h+1})_U$ of $g_h(z)$ along the classifying map $U \rightarrow B_{\text{ét}}C_p$. This is the scalar denoted by g_h in derived quotients. Changing the lift by an element of $J_h[[Z]]$ leaves the quotient vanishing test unchanged.

The constant-cochain map and the norm, followed by the ordinary transfer quotient, define two ring homomorphisms:

$$\iota, \overline{N}_{\widehat{\text{PMGL}}_{h+1}} : \pi_{0,0}\widehat{\text{PMGL}}_{h+1} \longrightarrow \pi_{0,0}(\widehat{\text{PMGL}}_{h+1}^{B_{\text{ét}}C_p})/(\text{Tr}_{\widehat{\text{PMGL}}_{h+1}}(1)).$$

Here ι is the constant-cochain map followed by the quotient; the overbar on N denotes the same quotient on its target. Let F_ι and $F_{\overline{N}}$ be the base changes of F along these two maps. Part (i) below concerns formal groups over this transfer quotient. Part (ii) uses the original norm $N_{\widehat{\text{PMGL}}_{h+1}(R)}$, followed by a map to a derived quotient.

Proposition 3.4 (Geometric power operation). *The norm operations on the completed algebras in (3.2) have the following properties.*

- (i) *Let X denote the normalized Euler coordinate of the universal line bundle on $\mathbb{P}^\infty$. Its norm, followed by the transfer quotient, is a series*

$$H(X) \in \left(\pi_{0,0}(\widehat{\text{PMGL}}_{h+1}^{B_{\text{ét}}C_p})/(\text{Tr}_{\widehat{\text{PMGL}}_{h+1}}(1)) \right) [[X]].$$

This series defines a homomorphism from the formal group with law F_ι to the one with law $F_{\overline{N}}$: over the transfer quotient, the following square commutes:

$$\begin{array}{ccc} \widehat{\mathbb{A}}^1 \times \widehat{\mathbb{A}}^1 & \xrightarrow{H \times H} & \widehat{\mathbb{A}}^1 \times \widehat{\mathbb{A}}^1 \\ F_\iota \downarrow & & \downarrow F_{\overline{N}} \\ \widehat{\mathbb{A}}^1 & \xrightarrow{H} & \widehat{\mathbb{A}}^1 \end{array}$$

The series $H(X)$ has the following Euler expression, with the coefficients of F read through ι :

$$H(X) = V(X, z) \prod_{j=0}^{p-1} (X +_F [j]_F(z)), \quad V(X, z) \text{ a unit.}$$

- (ii) *For every $x \in \pi_{0,0}\widehat{\text{PMGL}}_{h+1}(R)$, a zero image in the source quotient implies a zero image in the following derived target:*

$$\begin{aligned} x = 0 \text{ in } \pi_{0,0}(\widehat{\text{PMGL}}_{h+1}(R)/(J_h, \beta)) \\ \implies N_{\widehat{\text{PMGL}}_{h+1}(R)}(x) = 0 \\ \text{in } \pi_{0,0}((\widehat{\text{PMGL}}_{h+1}(R))^{B_{\text{ét}}C_p}/(J_h, \beta, g_h)). \end{aligned} \tag{3.6}$$

For part (i), compare the norm of a Thom class with the class defined by the orientation. Part (ii) will follow from a section of the normed dual unit in Proposition 2.1.

Proof of Proposition 3.4(i). On a finite-dimensional approximation to the universal cyclic torsor $f : V \rightarrow U$, let L be a line bundle over U . Applying the relative $\widehat{\text{PMGL}}_{h+1}$ -norm to a Thom equivalence for f^*L gives

$$\Sigma^{f^*L}(\widehat{\text{PMGL}}_{h+1})_U \simeq \Sigma^{f^*\mathcal{O}_V}(\widehat{\text{PMGL}}_{h+1})_U.$$

The orientation gives another Thom equivalence between the same modules. The ratio of the two Thom classes is a unit, possibly depending on L . The norm of the zero section is the zero section of $f_* f^* L$ [7, Proposition 3.13]. After decomposing the regular representation into characters, the Euler product follows, with the unit $V(X, z)$ of Proposition 3.4(i).

Take two universal line bundles with Euler coordinates X and Y . The norm of the Euler class of their tensor product is $H(F_\ell(X, Y))$. Applying the norm to the coefficients of F and to the Euler classes X, Y expresses the same class as $F_{\overline{N}}(H(X), H(Y))$. Thus

$$H(F_\ell(X, Y)) = F_{\overline{N}}(H(X), H(Y)). \quad (3.7)$$

On each finite projective-bundle approximation, naturality under tensor product of line bundles proves the identity to the corresponding finite order. The Euler product implies $H(X) \in (X)$, so substitution preserves the coordinate-adic filtration. The inverse limit proves (3.7) for formal series. The relative module norms are constructed in [7, Propositions 7.6(4) and 7.17]. $\square$

3.3 Coefficient families that split the Bockstein maps

We first prove that the norm kills the lower height parameters in an ordinary coefficient quotient. This calculation will trivialize the monodromies of the lines whose Thom objects represent the Bockstein boundaries.

Lemma 3.5. *The class z is regular in $((\pi_{0,0}\widehat{\text{PMGL}}_{h+1})/J_h)[[z]]/(g_h)$, and $\overline{N}_{\widehat{\text{PMGL}}_{h+1}}(a_i) = 0$ in this ring for $0 \leq i < h$.*

Proof. Multiplication by z is injective: if $zf = g_h b$, evaluation at zero and regularity of $g_h(0) = a_h$ give $b(0) = 0$; writing $b = zc$ and cancelling z proves the claim. Thus $H'(0) = z^{p-1} \cdot \text{unit}$ is regular in this ring. The transfer formula [6, Lemma 4.12] gives

$$\text{Tr}_{\widehat{\text{PMGL}}_{h+1}}(1) = \text{unit} \cdot [p]_F(z)/z.$$

The transfer class vanishes modulo (J_h, g_h) because $[p]_F(z)/z = z^{p^h-1}g_h$ modulo J_h . Hence the coefficient quotient map factors as

$$\pi_{0,0}(\widehat{\text{PMGL}}_{h+1}^{B_{\text{tors}}C_p})/(\text{Tr}_{\widehat{\text{PMGL}}_{h+1}}(1)) \longrightarrow ((\pi_{0,0}\widehat{\text{PMGL}}_{h+1})/J_h)[[z]]/(g_h).$$

The values of $\overline{N}_{\widehat{\text{PMGL}}_{h+1}}$ in this lemma mean its composite with this map. The composite is additive and unital; in particular, $\overline{N}_{\widehat{\text{PMGL}}_{h+1}}(p) = 0$. The source p -series has no term below degree p^h . Inductively comparing the X^{p^i} coefficients in the formal-group homomorphism identity of Proposition 3.4(i) therefore gives

$$\overline{N}_{\widehat{\text{PMGL}}_{h+1}}(a_i)H'(0)^{p^i} = 0 \quad (1 \leq i < h).$$

Cancellation proves the result. $\square$

Recall the structure map $\pi : B_{\text{tors}}C_p \rightarrow \text{Spec } k$. The next lemma gives free relative cells after the fixed diagonal has been included. The obstruction on each free cell is then evaluated at the trivial torsor.

Lemma 3.6 (Relative cells in permuted sphere products). *Let C_p act on $(S^r)^p$ by cyclic permutation. The following inclusions admit relative equivariant CW approximations with only free cells:*

- (i) *The union of the diagonal and the coordinate circles in $(S^1)^p$, with relative cells of dimensions at least 2.*
- (ii) *The diagonal S^3 in $(S^3)^p$, with relative cells of dimensions at least 3.*

More precisely, writing the relevant inclusion as $W \hookrightarrow X$, there is a factorization $W \hookrightarrow X' \rightarrow X$ under W such that the first map attaches cells $C_p \times D^d$ in the indicated dimensions and the second is a weak equivalence on all fixed-point spaces.

Proof. In both cases $W^{C_p} = X^{C_p}$ is the diagonal. In (i), W is connected and its coordinate circles generate $\pi_1((S^1)^p)$; thus $W \rightarrow X$ is 1-connected. In (ii), both W and X are 2-connected, so $W \rightarrow X$ is 2-connected.

Apply the relative CW approximation construction equivariantly, starting in dimension 2 or 3, respectively. At each stage, represent the next relative homotopy classes by maps of disks to X with boundary in the preceding stage. Attach a free C_p -orbit of disks for each such map, and extend to the other disks by translation. The ordinary relative homotopy argument increases the connectivity at each stage; its colimit $X' \rightarrow X$ is therefore an underlying weak equivalence. Free cells introduce no new fixed points, so $(X')^{C_p} = W^{C_p} \rightarrow X^{C_p}$ remains an equivalence. Since p is prime, the trivial subgroup and C_p are the only subgroups to test. This proves the asserted equivariant weak equivalence under W . $\square$

Lemma 3.7. *Let*

$$Q \in \text{CAlg}_{\pi^* \widehat{\text{PMGL}}_{h+1}}(\text{SH}(B_{\text{tors}} C_p)).$$

Write Q_f for its evaluation at a torsor $f : V \rightarrow U$, and Q_{triv} for its evaluation at the trivial torsor over k . Suppose

$$\pi_{j,0} Q_{\text{triv}} = 0 \quad (j \neq 0), \quad \pi_2 \Gamma(B_{\text{tors}} C_p; Q) = 0, \quad (3.8)$$

and, for $0 \leq i < h$, the images of the indicated classes satisfy

$$\begin{aligned} a_i &= 0 \quad \text{in } \pi_{0,0} Q_{\text{triv}}, \\ N_{\widehat{\text{PMGL}}_{h+1}}(1 + a_i) &= 1 \quad \text{in } \pi_0 \Gamma(B_{\text{tors}} C_p; Q). \end{aligned} \quad (3.9)$$

For $b = a_i$ take $d = 0$, and for $b = \beta$ take $d = 2$. Let $\partial_b : (\widehat{\text{PMGL}}_{h+1})_V / b \rightarrow \Sigma^{d+1,0}(\widehat{\text{PMGL}}_{h+1})_V$ be the specified Bockstein boundary. For each torsor $f : V \rightarrow U$, form the Q_f -linear arrow

$$\begin{aligned} \partial_{b,f}^Q : \quad & Q_f \otimes_{(\widehat{\text{PMGL}}_{h+1})_U} N_f^{\widehat{\text{PMGL}}_{h+1}}((\widehat{\text{PMGL}}_{h+1})_V / b) \\ & \xrightarrow{1 \otimes N_f^{\widehat{\text{PMGL}}_{h+1}}(\partial_b)} Q_f \otimes_{(\widehat{\text{PMGL}}_{h+1})_U} N_f^{\widehat{\text{PMGL}}_{h+1}}(\Sigma^{d+1,0}(\widehat{\text{PMGL}}_{h+1})_V). \end{aligned}$$

These arrows admit sections $s_{b,f}$ with $\partial_{b,f}^Q s_{b,f} \simeq \text{id}$, coherently compatible with pullback of torsors. Equivalently, norm base change assembles the arrows into a morphism in $\text{Mod}_Q(\text{SH}(B_{\text{tors}} C_p))$, and this morphism admits a section in that category.

Proof. The two line diagrams below, on S^1 and S^3 , both have fibre $\widehat{\text{PMGL}}_{h+1}$ at the chosen vertex. Their norms, after extension to Q , have base-point fibre Q . They therefore take values in the component of the unit module and have classifying maps

$$\ell_Q : \text{Assoc}_{C_p}(X) \longrightarrow B\text{GL}_1(Q) \quad \text{over } B_{\text{tors}}C_p,$$

where $X = (S^r)^p$ and $r \in \{1, 3\}$, using (2.6). A based trivialization means a homotopy from ℓ_Q to the map classifying the constant line Q , fixed at the chosen vertex. By Lemma 2.2, such a homotopy gives source and target equivalences identifying the normed boundary with the relative collapse for the constant line. We first construct the homotopy, then use these two equivalences to specify the section.

Extension across free cells. Suppose a based trivialization of ℓ_Q is already given on $\text{Assoc}_{C_p}(W)$, where $W \subset X$ contains the fixed points and the relative cells are free of dimensions at least 2. For a characteristic map $C_p \times D^d \rightarrow X$, the orbit-presheaf adjunction identifies the restricted line with a map

$$D^d \longrightarrow B\text{GL}_1(\Gamma(k, Q_{\text{triv}})).$$

On $D^d \times \{0\}$ use this map, on $D^d \times \{1\}$ use the constant map, and on $\partial D^d \times [0, 1]$ use the previously chosen trivialization. They give a map from $\partial(D^d \times [0, 1]) \simeq S^d$ to $B\text{GL}_1(\Gamma(k, Q_{\text{triv}}))$. Its homotopy class is the extension obstruction, and it vanishes because

$$\pi_d B\text{GL}_1(\Gamma(k, Q_{\text{triv}})) = \pi_{d-1,0} Q_{\text{triv}} = 0 \quad (d \geq 2).$$

A nullhomotopy fills $D^d \times [0, 1]$ and extends the trivialization. Successive fillings give a homotopy on the relative CW approximation and hence on $\text{Assoc}_{C_p}(X)$. Because this is a homotopy of presheaf maps over $B_{\text{tors}}C_p$, its evaluations commute coherently with pullback of torsors.

The a_i boundary. Since a_i belongs to the completion ideal, $1 + a_i$ is a unit. The automorphism $1 + a_i$ defines a based map

$$\ell_{a_i} : S^1 = B\mathbb{Z} \longrightarrow B\text{GL}_1(\Gamma(k, \widehat{\text{PMGL}}_{h+1})).$$

Its Thom colimit is

$$\text{coeq}(1, 1 + a_i : \widehat{\text{PMGL}}_{h+1} \rightrightarrows \widehat{\text{PMGL}}_{h+1}) \simeq \widehat{\text{PMGL}}_{h+1}/a_i.$$

The relative Thom quotient for the chosen vertex is the specified boundary $\partial_{a_i} : \widehat{\text{PMGL}}_{h+1}/a_i \rightarrow \Sigma^{1,0}\widehat{\text{PMGL}}_{h+1}$.

On $(S^1)^p$, take W to be the union of the fixed diagonal and the coordinate circles. On the fixed diagonal, ℓ_Q corresponds to a based map

$$S^1 \longrightarrow \Gamma(B_{\text{tors}}C_p; B\text{GL}_1(Q)).$$

Its loop class is $N_{\widehat{\text{PMGL}}_{h+1}}(1 + a_i)$, so (2.8) and the second equality in (3.9) give a based nullhomotopy. The coordinate circles form one free orbit; their restriction is the map $S^1 \rightarrow B\text{GL}_1(\Gamma(k, Q_{\text{triv}}))$ with loop class $1 + a_i$. The first equality in (3.9) gives a based nullhomotopy of this map. The nullhomotopies agree at the common vertex and thus trivialize the line on W . Lemma 3.6(i) gives free relative cells of dimensions at least 2, so the preceding extension argument trivializes the line on $(S^1)^p$.

The β boundary. The clutching map $S^3 \rightarrow B\mathrm{GL}_1(\Gamma(k, \widehat{\mathrm{PMGL}}_{h+1}))$ representing β has hemisphere presentation

$$\widehat{\mathrm{PMGL}}_{h+1} \xleftarrow{(1,\beta)} \widehat{\mathrm{PMGL}}_{h+1} \oplus \Sigma^{2,0}\widehat{\mathrm{PMGL}}_{h+1} \xrightarrow{(1,0)} \widehat{\mathrm{PMGL}}_{h+1}.$$

The homotopy pushout is $\widehat{\mathrm{PMGL}}_{h+1}/\beta$, and collapse of the chosen hemisphere gives its specified boundary to $\Sigma^{3,0}\widehat{\mathrm{PMGL}}_{h+1}$.

On $(S^3)^p$, take W to be the fixed diagonal. The restriction of the normed line to W defines a based class in

$$\pi_3\Gamma(B_{\mathrm{tors}}C_p; B\mathrm{GL}_1(Q)) = \pi_2\Gamma(B_{\mathrm{tors}}C_p; Q) = 0.$$

The identification of π_3 with π_2 follows from right- Q -linear endomorphisms, as in Section 2. Choose a based nullhomotopy on W . By Lemma 3.6(ii), the relative cells outside W are free and have dimensions at least 3. Extending over these cells trivializes the line on $(S^3)^p$.

Sections of the specified arrows. Let $B \subset S^r$ be the chosen vertex for $r = 1$ or hemisphere for $r = 3$. Define the constant-line quotient map

$$\rho_r : \widehat{\mathrm{PMGL}}_{h+1} \otimes \Sigma_+^\infty S^r \longrightarrow \widehat{\mathrm{PMGL}}_{h+1} \otimes \Sigma^\infty(S^r/B) \simeq \Sigma^{r,0}\widehat{\mathrm{PMGL}}_{h+1}.$$

The inclusion from $\widehat{\mathrm{PMGL}}_{h+1} \otimes \Sigma_+^\infty B$ into its source has a retraction: compose the map induced by $S^r \rightarrow *$ with the inverse of the equivalence induced by $B \rightarrow *$. Hence ρ_r has a module section t_r with $\rho_r t_r \simeq \mathrm{id}$.

Lemma 2.2 identifies $N_f^{\widehat{\mathrm{PMGL}}_{h+1}}(\rho_{r,V})$ with

$$\mathrm{Th}_{\widehat{\mathrm{PMGL}}_{h+1},U}((S^r)^p, \underline{(\widehat{\mathrm{PMGL}}_{h+1})_U}) \longrightarrow \mathrm{Th}_{\widehat{\mathrm{PMGL}}_{h+1},U}((S^r)^p, F_B; \underline{(\widehat{\mathrm{PMGL}}_{h+1})_U}),$$

where the underlined coefficient is the constant line on the associated presheaf, and $F_B = \bigcup_{j=1}^p (S^r)^{j-1} \times B \times (S^r)^{p-j}$. This is the constant-coefficient relative collapse.

Take $b = a_i$ when $r = 1$ and $b = \beta$ when $r = 3$. The chosen line trivialization on $(S^r)^p$, restricted also to F_B , gives source and target equivalences α_f and γ_f from $\partial_{b,f}^Q$ to $1 \otimes N_f^{\widehat{\mathrm{PMGL}}_{h+1}}(\rho_{r,V})$. Thus

$$\gamma_f \partial_{b,f}^Q = (1 \otimes N_f^{\widehat{\mathrm{PMGL}}_{h+1}}(\rho_{r,V})) \alpha_f.$$

Since the norm preserves the identity $\rho_r t_r \simeq \mathrm{id}$, the required section is

$$s_{b,f} = \alpha_f^{-1} (1 \otimes N_f^{\widehat{\mathrm{PMGL}}_{h+1}}(t_{r,V})) \gamma_f.$$

The norm comparison and the trivialization commute with pullback, so these maps form a section in $\mathrm{Mod}_Q(\mathrm{SH}(B_{\mathrm{tors}}C_p))$. The construction uses the norm on $\widehat{\mathrm{PMGL}}_{h+1}$ and retains its normed suspension as target. $\square$

Proof of Proposition 3.4(ii). In $\mathrm{Mod}_{\pi^*\widehat{\mathrm{PMGL}}_{h+1}}(\mathrm{SH}(B_{\mathrm{tors}}C_p))$, form the finite scalar quotient

$$Q := \pi^*\widehat{\mathrm{PMGL}}_{h+1}/(J_h, \beta, g_h).$$

Evaluating the chosen scalar g_h at a torsor $f : V \rightarrow U$ gives $g_h(f)$, so

$$Q_f = (\widehat{\mathrm{PMGL}}_{h+1})_U/(J_h, \beta, g_h(f)).$$

Apply Lemma 2.3 in $\text{Mod}_{\pi^*\widehat{\text{PMGL}}_{h+1}}(\text{SH}(B_{\text{tors}}C_p))$, whose unit is $\pi^*\widehat{\text{PMGL}}_{h+1}$. The lemma gives Q a commutative $\pi^*\widehat{\text{PMGL}}_{h+1}$ -algebra structure with the quotient unit.

The parameters in J_h form a regular sequence on the cochain coefficient ring, which is finite free over $\pi_{0,0}\widehat{\text{PMGL}}_{h+1}$. Quotienting by J_h and β leaves a discrete section spectrum. Taking the final g_h -cofiber adds only its kernel in degree one. Consequently,

$$\begin{aligned}\pi_j\Gamma(B_{\text{tors}}C_p; Q) &= 0 & (j \notin \{0, 1\}), \\ \pi_0\Gamma(B_{\text{tors}}C_p; Q) &= ((\pi_{0,0}\widehat{\text{PMGL}}_{h+1})/J_h)[[z]]/(g_h).\end{aligned}$$

At the trivial torsor, the last scalar is a_h modulo J_h . Regularity of a_h on $(\pi_{0,0}\widehat{\text{PMGL}}_{h+1})/J_h$ implies that $\Gamma(k, Q_{\text{triv}})$ is discrete. This proves both conditions in (3.8). Moreover, $a_i \in J_h$ proves the first identity in (3.9). Lemma 3.5, together with additivity modulo transfer, proves the second identity. Lemma 3.7 therefore splits the normed a_i - and β -boundaries in $\text{Mod}_Q(\text{SH}(B_{\text{tors}}C_p))$.

For $b \in \{a_0, \dots, a_{h-1}, \beta\}$, write $b : \Sigma^{d,0}\widehat{\text{PMGL}}_{h+1} \rightarrow \widehat{\text{PMGL}}_{h+1}$, where $d = 0$ for a_i and $d = 2$ for β . Let $c_b : \widehat{\text{PMGL}}_{h+1} \rightarrow \widehat{\text{PMGL}}_{h+1}/b$ be the quotient unit. The equivalence

$$(\widehat{\text{PMGL}}_{h+1}/b)^\vee \simeq \Sigma^{-d-1,0}(\widehat{\text{PMGL}}_{h+1}/b)$$

identifies $c_b^\vee$ with $\Sigma^{-d-1,0}\partial_b$. For each torsor $f : V \rightarrow U$, tensor $\partial_{b,f}^Q$ and its section with the inverse of the invertible module $N_f^{\widehat{\text{PMGL}}_{h+1}}(\Sigma^{d+1,0}(\widehat{\text{PMGL}}_{h+1})_V)$. By the symmetric monoidality of $N_f^{\widehat{\text{PMGL}}_{h+1}}$, the resulting split arrow is

$$Q_f \otimes_{(\widehat{\text{PMGL}}_{h+1})_U} N_f^{\widehat{\text{PMGL}}_{h+1}}(((\widehat{\text{PMGL}}_{h+1})_V/b)^\vee) \xrightarrow{1 \otimes N_f^{\widehat{\text{PMGL}}_{h+1}}(c_b^\vee)} Q_f.$$

The quotient $\widehat{\text{PMGL}}_{h+1}/(J_h, \beta)$ is the tensor product of the scalar cofibers for the generators of J_h and for β . Tensoring their sections gives a section of the arrow whose value at f is

$$N_f^{\widehat{\text{PMGL}}_{h+1}}(((\widehat{\text{PMGL}}_{h+1})_V/(J_h, \beta))^\vee) \otimes_{(\widehat{\text{PMGL}}_{h+1})_U} Q_f \longrightarrow Q_f.$$

The tensor products are taken in $\text{Mod}_Q(\text{SH}(B_{\text{tors}}C_p))$, so the sections commute coherently with pullback.

Exactness of $\Gamma(B_{\text{tors}}C_p; -)$ allows us to take the (J_h, β, g_h) -quotient before or after taking sections. Together with (2.3), this identifies

$$\pi_0\Gamma(B_{\text{tors}}C_p; \pi^*(\widehat{\text{PMGL}}_{h+1}(R)) \otimes_{\pi^*\widehat{\text{PMGL}}_{h+1}} Q) \cong \pi_{0,0}((\widehat{\text{PMGL}}_{h+1}(R))^{B_{\text{et}}C_p}/(J_h, \beta, g_h)).$$

Apply Proposition 2.1 to the quotient unit $\widehat{\text{PMGL}}_{h+1} \rightarrow \widehat{\text{PMGL}}_{h+1}/(J_h, \beta)$ and the section just constructed over Q . The section-spectrum identification proves the norm vanishing asserted in (3.6). $\square$

3.4 Coefficient projections and exponent decrease

Let $d = p^h(p-1)$. The coefficient ring $(\pi_{0,0}\widehat{\text{PMGL}}_{h+1})/J_h$ is complete in a_h . The constant coefficient of g_h is a_h , and modulo a_h its first nonzero term is a unit times z^d . The formal implicit-function theorem therefore gives, in $((\pi_{0,0}\widehat{\text{PMGL}}_{h+1})/J_h)[[z]]/(g_h)$,

$$a_h = z^d \varepsilon(z), \quad \overline{N}_{\widehat{\text{PMGL}}_{h+1}}(a_h) = z^{p-1} \delta(z), \quad (3.10)$$

where both factors are units. For the second identity, the Euler expression gives

$$H'(0) = V(0, z) \prod_{j=1}^{p-1} [j]_F(z) = z^{p-1} \cdot \text{unit}.$$

Lemma 3.5 gives

$$\overline{N}_{\widehat{\text{PMGL}}_{h+1}}(a_i) = 0 \quad (i < h).$$

The same lemma proves that z is regular in $((\pi_{0,0}\widehat{\text{PMGL}}_{h+1})/J_h)[[z]]/(g_h)$. The X^{p^h} coefficient in the formal-group homomorphism identity therefore gives

$$H'(0)a_h = \overline{N}_{\widehat{\text{PMGL}}_{h+1}}(a_h)H'(0)^{p^h}.$$

Substitute $a_h = z^d \varepsilon(z)$ and $H'(0) = z^{p-1} \cdot \text{unit}$. Cancelling the common power of the regular element z proves the formula for $\overline{N}_{\widehat{\text{PMGL}}_{h+1}}(a_h)$ in (3.10).

The coefficient projections must be induced by $\widehat{\text{PMGL}}_{h+1}$ -module maps

$$\widehat{\text{PMGL}}_{h+1}^{B_{\text{ét}}C_p} / (J_h, \beta, g_h) \longrightarrow \widehat{\text{PMGL}}_{h+1} / (J_h, \beta).$$

The finite-free description of $\widehat{\text{PMGL}}_{h+1}^{B_{\text{ét}}C_p}$ shows that both source and target lie in the thick subcategory generated by ordinary suspensions of $\widehat{\text{PMGL}}_{h+1}$. The sections of $\widehat{\text{PMGL}}_{h+1} / (J_h, \beta)$ form $H((\pi_{0,0}\widehat{\text{PMGL}}_{h+1})/J_h)$. For the source, taking the J_h - and β -quotients also leaves discrete sections. The final g_h -cofiber has only the cokernel of g_h in degree zero and its kernel in degree one. In particular, the degree-zero coefficient ring of the source is $((\pi_{0,0}\widehat{\text{PMGL}}_{h+1})/J_h)[[z]]/(g_h)$.

Weierstrass preparation makes the degree-zero ring free over $(\pi_{0,0}\widehat{\text{PMGL}}_{h+1})/J_h$ with basis $1, z, \dots, z^{d-1}$. For $0 \leq s < d$, let

$$\lambda_s : ((\pi_{0,0}\widehat{\text{PMGL}}_{h+1})/J_h)[[z]]/(g_h) \longrightarrow (\pi_{0,0}\widehat{\text{PMGL}}_{h+1})/J_h, \quad \sum_{j=0}^{d-1} c_j z^j \longmapsto c_s.$$

These are linear maps for the constant-coefficient action, not ring homomorphisms. Lemma 2.4 gives $\widehat{\text{PMGL}}_{h+1}$ -module maps

$$\text{pr}_s : \widehat{\text{PMGL}}_{h+1}^{B_{\text{ét}}C_p} / (J_h, \beta, g_h) \longrightarrow \widehat{\text{PMGL}}_{h+1} / (J_h, \beta)$$

with $\pi_{0,0} \text{pr}_s = \lambda_s$. The map pr_s is obtained by applying $\widehat{\text{PMGL}}_{h+1} \otimes_{\Gamma(k, \widehat{\text{PMGL}}_{h+1})} -$ to the composite of the degree-zero truncation of the source section spectrum and $H\lambda_s$, and then using the counit equivalences of that lemma. The truncation map exists even when the degree-one kernel of g_h is nonzero.

The completed homology calculation [6, Lemma 4.9], applied to these periodic coefficients, gives a finite free $\widehat{\text{PMGL}}_{h+1}$ -module

$$(\widehat{\text{PMGL}}_{h+1} \wedge \Sigma_+^\infty B_{\text{ét}}C_p)_{J_{h+1}}^\wedge \simeq \bigoplus_{j=0}^{p^{h+1}-1} \widehat{\text{PMGL}}_{h+1};$$

the powers of u identify the Tate suspensions with the displayed copies. For a J_{h+1} -complete $\widehat{\text{PMGL}}_{h+1}$ -module M , the internal $\widehat{\text{PMGL}}_{h+1}$ -linear Hom from this completed module to M is

$M^{B_{\text{ét}}C_p}$. Its $\widehat{\text{PMGL}}_{h+1}$ -linear dual is $\widehat{\text{PMGL}}_{h+1}^{B_{\text{ét}}C_p}$, so duality identifies the natural evaluation comparison as an equivalence

$$\widehat{\text{PMGL}}_{h+1}(R) \otimes_{\widehat{\text{PMGL}}_{h+1}} \widehat{\text{PMGL}}_{h+1}^{B_{\text{ét}}C_p} \xrightarrow{\simeq} (\widehat{\text{PMGL}}_{h+1}(R))^{B_{\text{ét}}C_p}.$$

Here the comparison is adjoint to evaluation on $\Sigma_+^\infty B_{\text{ét}}C_p$, and $\widehat{\text{PMGL}}_{h+1}(R)$ is complete by definition. Tensoring pr_s with $\widehat{\text{PMGL}}_{h+1}(R)$ therefore gives the maps

$$\text{pr}_{s,R} : (\widehat{\text{PMGL}}_{h+1}(R))^{B_{\text{ét}}C_p} / (J_h, \beta, g_h) \longrightarrow \widehat{\text{PMGL}}_{h+1}(R) / (J_h, \beta).$$

They extract the chosen coefficient from images of coefficient classes in (3.6). This uses no cellularity assumption on R or $\widehat{\text{PMGL}}_{h+1}(R)$.

Proposition 3.8. *If a_h has nilpotent image in the coefficient ring*

$$\pi_{0,0}(\widehat{\text{PMGL}}_{h+1}(R) / (J_h, \beta)),$$

then $\widehat{\text{PMGL}}_{h+1}(R) / (J_h, \beta) \simeq 0$.

Proof. We prove the exponent-reduction implication

$$a_h^N = 0 \implies a_h^{\lfloor N/p^h \rfloor} = 0 \quad \text{in } \pi_{0,0}(\widehat{\text{PMGL}}_{h+1}(R) / (J_h, \beta)).$$

Suppose $a_h^N = 0$ for $N > 0$. By (3.6) and multiplicativity, $N_{\widehat{\text{PMGL}}_{h+1}(R)}(a_h)^N = 0$ in the derived target. Naturality along $\widehat{\text{PMGL}}_{h+1} \rightarrow \widehat{\text{PMGL}}_{h+1}(R)$ identifies the norm class with the image of $N_{\widehat{\text{PMGL}}_{h+1}}(a_h)^N$. We compute its basis coefficients in the ordinary quotient, where the norm class is represented by $\overline{N}_{\widehat{\text{PMGL}}_{h+1}}(a_h)^N$. Write

$$N(p-1) = qd + s, \quad 0 \leq s < d, \quad q = \lfloor N/p^h \rfloor.$$

In the ordinary coefficient quotient,

$$\overline{N}_{\widehat{\text{PMGL}}_{h+1}}(a_h)^N = a_h^q z^s \varepsilon(z)^{-q} \delta(z)^N.$$

Modulo a_h the coefficient quotient is

$$((\pi_{0,0}\widehat{\text{PMGL}}_{h+1}) / J_{h+1})[z] / z^d.$$

The z^s -coefficient of $z^s \varepsilon(z)^{-q} \delta(z)^N$ is a unit modulo a_h , hence a unit in the a_h -complete ring $(\pi_{0,0}\widehat{\text{PMGL}}_{h+1}) / J_h$. Apply $\text{pr}_{s,R}$ to the norm relation. By its construction from λ_s , the image of the norm class is a_h^q times the image of that unit in $\pi_{0,0}(\widehat{\text{PMGL}}_{h+1}(R) / (J_h, \beta))$. It is zero because the norm class is zero. Cancelling the unit proves

$$a_h^N = 0 \implies a_h^{\lfloor N/p^h \rfloor} = 0 \quad \text{in } \pi_{0,0}(\widehat{\text{PMGL}}_{h+1}(R) / (J_h, \beta)). \quad (3.11)$$

Choose the least $N \geq 0$ such that a_h^N has zero image in $\pi_{0,0}(\widehat{\text{PMGL}}_{h+1}(R) / (J_h, \beta))$. If $N > 0$, then $\lfloor N/p^h \rfloor < N$, contradicting minimality by (3.11). Thus $N = 0$, and the quotient unit of the commutative algebra $\widehat{\text{PMGL}}_{h+1}(R) \otimes_{\widehat{\text{PMGL}}_{h+1}} \widehat{\text{PMGL}}_{h+1} / (J_h, \beta)$ is null. Hence

$$\widehat{\text{PMGL}}_{h+1}(R) / (J_h, \beta) \simeq 0. \quad (3.12)$$

□

3.5 The Bott-periodic quotient

Proof of Theorem 3.1. The relation $a_h^N = 0$ from Lemma 3.3 remains true after quotienting by β . Proposition 3.8 therefore implies

$$\widehat{\text{PMGL}}_{h+1}(R)/(J_h, \beta) \simeq 0.$$

We prove that $\widehat{\text{PMGL}}_{h+1}(R)/J_{h+1}$ is hyperétale and p -complete, and that its hyperétale realization vanishes. Rigidity will imply $\widehat{\text{PMGL}}_{h+1}(R)/J_{h+1} = 0$; taking the rank-zero summand will then prove (3.1).

Bott periodicity and p -completeness. The I_{h+1} -completion map induces an equivalence after taking the finite quotient by I_{h+1} , or equivalently by J_{h+1} :

$$\widehat{\text{PMGL}}_{h+1}(R)/J_{h+1} \simeq (\text{PMGL}_{(p)}[v_{h+1}^{-1}]/I_{h+1}) \wedge R. \quad (3.13)$$

For one parameter, multiplication by that parameter is invertible on the fiber of the completion map [7, Lemma 12.13]. Tensoring the map with the scalar cofiber therefore gives an equivalence. Each scalar cofiber is dualizable as a module over $\text{PMGL}_{(p)}[v_{h+1}^{-1}]$, so relative tensoring with it commutes with the limits defining the remaining completions. The diagonal in the multi-indexed inverse system is coinital, so these successive completions agree with the completion used in (3.13). Iterate over the generators of I_{h+1} and use the periodic unit to identify the I_{h+1} - and J_{h+1} -quotients. This proves (3.13). We can therefore compute the realization of $\widehat{\text{PMGL}}_{h+1}(R)/J_{h+1}$ from the finite quotient on the right-hand side.

Since $J_{h+1} = (J_h, a_h)$, commuting the finite scalar cofibers gives

$$(\widehat{\text{PMGL}}_{h+1}(R)/J_{h+1})/\beta \simeq (\widehat{\text{PMGL}}_{h+1}(R)/(J_h, \beta))/a_h \simeq 0.$$

By (3.12), multiplication by β on $\widehat{\text{PMGL}}_{h+1}(R)/J_{h+1}$ is therefore an equivalence. The equality $\tau = u\beta$ and invertibility of u imply that multiplication by τ is an equivalence as well.

The quotient $\widehat{\text{PMGL}}_{h+1}(R)/J_{h+1}$ contains a scalar p -cofiber, so multiplication by p^2 is null. For any p -inverted spectrum T , multiplication by p on

$$\text{Map}(T, \widehat{\text{PMGL}}_{h+1}(R)/J_{h+1})$$

is invertible by precomposition and nilpotent by postcomposition. The mapping spectrum is zero. Thus $\widehat{\text{PMGL}}_{h+1}(R)/J_{h+1}$ is orthogonal to every p -inverted spectrum and is derived p -complete.

To apply Bott inversion, we compare τ with the spherical Bott element. The map $\text{MGL}_p^\wedge \rightarrow \text{HZ}_p^\wedge$ is an isomorphism on $\pi_{0,-1}$, and the spherical Bott element maps to a primitive class in $\pi_{0,-1}(\text{HZ}_p^\wedge)$. Thus the coefficient class τ agrees, up to a unit, with the image of the spherical Bott element.

Since k is separably closed of characteristic different from p , the Bott-inversion theorem applies to the τ -invertible, p -complete spectrum $\widehat{\text{PMGL}}_{h+1}(R)/J_{h+1}$ and shows that it is hyperétale [4, Proposition 4.7 and Corollary 7.6(3)]. Rigidity identifies the hyperétale p -complete category over k with $\text{Sp}_p^\wedge$ [2, Theorem 3.1]. It therefore suffices to prove that the realization of $\widehat{\text{PMGL}}_{h+1}(R)/J_{h+1}$ is zero.

Vanishing after realization. Let $\text{SH}_{\text{hét}}(k)$ denote the hyperétale motivic category. Define e_k by the composite

$$\text{SH}(k) \xrightarrow{L_{\text{hét}}} \text{SH}_{\text{hét}}(k) \xrightarrow{(-)_p^\wedge} \text{SH}_{\text{hét}}(k)_p^\wedge \xrightarrow{\simeq} \text{Sp}_p^\wedge,$$

where the last arrow is the inverse of the constant-sheaf rigidity equivalence over k . The tensor product in $\mathrm{Sp}_p^\wedge$ is $(X \wedge Y)_p^\wedge$. Hyperétale localization, completion into the complete category, and the rigidity equivalence are colimit-preserving symmetric monoidal functors. Hence so is e_k . The oriented equivalence $e_k(\mathrm{MGL}_{(p)}) \simeq \mathrm{MU}_p^\wedge$ preserves the specified coefficient maps [3, Theorem 4.26, Proposition 4.31, and Corollary 4.32]. Since $\mathrm{Spec} k$ is strictly henselian, this equivalence has the $\mathbb{E}_\infty$ refinement compatible with the commutative coefficient structures. Use the isomorphism $\mathbb{Z}_p(1) \simeq \mathbb{Z}_p$ that sends the chosen compatible system of roots of unity to 1 to identify the Tate twists in realization. Applying e_k to the cofibers and telescopes in (1.1) then yields

$$e_k(K(j)) \simeq K^{\mathrm{top}}(j) \quad (j \geq 1).$$

The lax monoidal inclusion of p -complete spectra into spectra makes $e_k(R)$ an ordinary $\mathbb{E}_\infty$ -ring. Apply e_k to the hypothesis $K(h) \wedge R = 0$. The p -completion of $K^{\mathrm{top}}(h) \wedge e_k(R)$ is zero. Since $K^{\mathrm{top}}(h) \wedge e_k(R)$ is killed by a fixed power of p , the same argument with maps from p -inverted spectra shows that the smash product is already p -complete. Consequently,

$$K^{\mathrm{top}}(h) \wedge e_k(R) = 0, \quad K^{\mathrm{top}}(h+1) \wedge e_k(R) = 0,$$

where the second vanishing is Hahn's theorem [11, Theorems 1.1 and 1.3].

The proof of Lemma 3.2 also applies in ordinary spectra: use the classical Landweber coefficient formula and suspensions of the sphere as compact generators. The comparison of the two height-ideal actions, the Koszul argument, and the two-sided bar are unchanged. Applied to $e_k(R)$, this comparison turns $K^{\mathrm{top}}(h+1) \wedge e_k(R) = 0$ into

$$(\mathrm{MU}_{(p)}[v_{h+1}^{-1}]/I_{h+1}) \wedge e_k(R) = 0.$$

Here I_{h+1} acts through the first MU -factor. Apply realization to (3.13). The MU -linear rank decomposition of realized periodic cobordism expresses the realization of the right-hand side of (3.13) as the p -completed sum of the Σ^{2r} -suspensions of this zero spectrum, for $r \in \mathbb{Z}$. Consequently,

$$e_k(\widehat{\mathrm{PMGL}}_{h+1}(R)/J_{h+1}) = 0.$$

The spectrum $\widehat{\mathrm{PMGL}}_{h+1}(R)/J_{h+1}$ is hyperétale and p -complete by the first part of the proof. Since realization is an equivalence on this subcategory, the vanishing of its realization implies

$$\widehat{\mathrm{PMGL}}_{h+1}(R)/J_{h+1} = 0.$$

To deduce (3.1), we pass from periodic cobordism back to MGL by taking the rank-zero summand. First restore the homogeneous v_i in the MGL -module decomposition

$$\mathrm{PMGL} \simeq \bigoplus_{r \in \mathbb{Z}} \Sigma^{2r,r} \mathrm{MGL}.$$

The homogeneous v_i preserve rank summands; the normalized a_i need not. Taking the rank-zero retract of (3.13) therefore shows that the following spectrum vanishes:

$$(\mathrm{MGL}_{(p)}[v_{h+1}^{-1}]/I_{h+1}) \wedge R.$$

This proves (3.1). Take the further scalar cofibers for the generators t_d not already killed by I_{h+1} , with $d \neq p^{h+1} - 1$, and the filtered colimit prescribed in (1.1). The result is $K(h+1) \wedge R = 0$, which proves Theorem 3.1. $\square$

4 From fields to schemes

We first descend along separable field extensions, then from residue fields to schemes. In both arguments, dualizability allows us to express the identity of a module as a class with compact source. If this class vanishes over a limit of schemes, continuity makes it vanish at a finite stage.

Lemma 4.1. *Let $S_\infty = \lim_\alpha S_\alpha$ be an inverse limit of quasi-compact, quasi-separated schemes with affine transition maps. Fix a stage S_0 , an algebra $D \in \mathrm{CAlg}(\mathrm{SH}(S_0))$, and a dualizable D -module B . If $B_{S_\infty} \simeq 0$, then $B_{S_\alpha} \simeq 0$ at some stage over S_0 .*

Proof. The internal endomorphism D -module is $\underline{\mathrm{End}}_D(B) \simeq B^\vee \otimes_D B$. After forgetting to $\mathrm{SH}(S_0)$, the adjunction identifies

$$[\mathbf{1}_{S_0}, \underline{\mathrm{End}}_D(B)] \simeq [B, B]_{\mathrm{Mod}_D(\mathrm{SH}(S_0))}.$$

Let $\mathbf{1}_{S_0} \rightarrow \underline{\mathrm{End}}_D(B)$ represent the class corresponding to id_B . Since pullback preserves duals and relative tensor products, it carries this class to the identity of the pulled-back module. Continuity with compact source $\mathbf{1}_{S_0}$ and arbitrary fixed target [8, Appendix A.9 and A.10(1)] therefore gives

$$\mathrm{colim}_\alpha [\mathbf{1}_{S_\alpha}, \underline{\mathrm{End}}_{D_{S_\alpha}}(B_{S_\alpha})] \simeq [\mathbf{1}_{S_\infty}, \underline{\mathrm{End}}_{D_{S_\infty}}(B_{S_\infty})].$$

Since $B_{S_\infty} = 0$, its identity class vanishes. The isomorphism above implies that the identity of B_{S_α} vanishes for some α , so $B_{S_\alpha} = 0$. $\square$

A finite derived scalar quotient of an algebra is dualizable over that algebra: it is a tensor product of cofibers of maps between invertible shifts of the free rank-one module. We apply the lemma to these quotients.

Lemma 4.2. *Let $n \geq 1$, let k be a field of characteristic different from p , and let R be normed over k . If*

$$\left((\mathrm{MGL}_{k,(p)}[v_n^{-1}]/I_n) \wedge R \right)_{k^{\mathrm{sep}}} \simeq 0,$$

then $(\mathrm{MGL}_{k,(p)}[v_n^{-1}]/I_n) \wedge R \simeq 0$.

Proof. The finite quotient $(\mathrm{MGL}_{k,(p)}[v_n^{-1}]/I_n) \wedge R$ is dualizable over $\mathrm{MGL}_{k,(p)}[v_n^{-1}] \wedge R$. Apply Lemma 4.1 to the finite separable subextensions of k^{sep}/k . It gives a finite separable extension L/k over which this quotient is zero.

The I_n -completion of $\mathrm{MGL}_{k,(p)}[v_n^{-1}] \wedge R$ is normed by [6, Corollary 3.5]. Its I_n -quotient is $(\mathrm{MGL}_{k,(p)}[v_n^{-1}]/I_n) \wedge R$, by finite-quotient invariance of completion [7, Lemma 12.13]. For $f : \mathrm{Spec} L \rightarrow \mathrm{Spec} k$, the functor f^* has a left adjoint $f_\sharp$ and a right adjoint f_* . It therefore preserves both limits and cofibers and commutes with this completion. Over L , each quotient by $(p^m, v_1^m, \dots, v_{n-1}^m)$ has a finite filtration with subquotients given by shifts of the zero I_n -quotient. The inverse limit of these quotients, which is the completed algebra over L , is therefore zero. By normed descent [6, Lemma 5.4], the completed algebra is also zero over k . Taking its I_n -quotient gives $(\mathrm{MGL}_{k,(p)}[v_n^{-1}]/I_n) \wedge R = 0$. $\square$

Lemma 4.3. *Let S be a scheme, let $D \in \mathrm{CAlg}(\mathrm{SH}(S))$, and let B be a dualizable D -module. If $B_{\kappa(s)} \simeq 0$ for every point $s \in S$, then $B \simeq 0$.*

Proof. The claim is Zariski local, so let $S = \operatorname{Spec} A$. We construct a finite stratification by locally closed subschemes on which B vanishes, then recover $B = 0$ from localization triangles. For a prime $\mathfrak{q} \subset A$, write

$$\kappa(\mathfrak{q}) = \operatorname{colim}_{\substack{J \subset \mathfrak{q} \text{ f.g.} \\ f \notin \mathfrak{q}}} (A/J)[1/f].$$

By Lemma 4.1, B vanishes on some $\operatorname{Spec}((A/J)[1/f])$ containing $\mathfrak{q}$. The sets $V(J) \cap D(f)$ thus obtained cover $\operatorname{Spec} A$. They are open in its compact constructible topology [16, Tag 0901], so finitely many suffice.

Collect the denominators and the chosen finite lists of ideal generators from this cover. For each collected element a , refine every current stratum T into $T \times_S V(a)$ and $T \times_S D(a)$. There are finitely many refinements. On every nonempty final stratum, membership in each covering set is constant. Choose a point of that stratum and a covering member containing it. The stratum factors through this member as a scheme: the refinement has set its ideal generators to zero and inverted its denominator. Hence B vanishes on every final stratum.

At each refinement, with complementary immersions $i : Z \rightarrow T$ and $j : U \rightarrow T$, the localization triangle is

$$j_! j^* B_T \rightarrow B_T \rightarrow i_* i^* B_T.$$

Vanishing on Z and U therefore implies vanishing on T . Induction backward through the finite sequence of refinements gives $B = 0$ on $\operatorname{Spec} A$. An affine open cover proves the assertion for arbitrary S . $\square$

Proof of Theorem 1.1. The coefficient models commute with pullback, as do normed structures [5, Corollary B.4]. For each $s \in S$, pull the hypothesis to $\kappa(s)^{\operatorname{sep}}$ and apply Theorem 3.1. Lemma 4.2 then gives

$$\left((\operatorname{MGL}_{S,(p)}[v_{h+1}^{-1}]/I_{h+1}) \wedge R \right)_{\kappa(s)} \simeq 0.$$

This finite quotient is dualizable over $\operatorname{MGL}_{S,(p)}[v_{h+1}^{-1}] \wedge R$. Lemma 4.3 therefore proves

$$(\operatorname{MGL}_{S,(p)}[v_{h+1}^{-1}]/I_{h+1}) \wedge R = 0.$$

The remaining coefficient quotients in (1.1) give $K_S(h+1) \wedge R = 0$. $\square$

5 The height-zero case

Proposition 5.1. *Let S be any scheme on which p is invertible and let $R \in \operatorname{NAlg}(\operatorname{SH}(S))$. If $\operatorname{H}\mathbb{Q}_S \wedge R \simeq 0$, then, for every $n \geq 1$,*

$$(\operatorname{MGL}_{S,(p)}[v_n^{-1}]/I_n) \wedge R \simeq 0, \quad K_S(n) \wedge R \simeq 0.$$

Proof. First let $S = \operatorname{Spec} k$ with k separably closed of characteristic different from p . Rational algebraic cobordism is a sum of Tate shifts of rational motivic cohomology [14, Corollary 10.6(i)]. It follows that $(\operatorname{MGL}_{(p)} \wedge R)[1/p] = 0$. The localization is the telescope

$$\operatorname{MGL}_{(p)} \wedge R \xrightarrow{p} \operatorname{MGL}_{(p)} \wedge R \xrightarrow{p} \cdots.$$

Compactness of the sphere implies that the class of the unit becomes zero at a finite stage of this telescope. Thus, for some N ,

$$p^N = 0 \quad \text{in } \pi_{0,0}(\operatorname{MGL}_{(p)} \wedge R).$$

For each n , the same relation holds in the normed algebra

$$(\mathrm{MGL}_{(p)}[v_n^{-1}])_{I_n}^\wedge \wedge R.$$

Since k is separably closed, the coefficient calculation of [6, §4] applies. Corollary 4.14 of that paper, applied to the relation $p^N = 0$, proves that this normed algebra is zero. Its finite I_n -quotient is $(\mathrm{MGL}_{(p)}[v_n^{-1}]/I_n) \wedge R$, by invariance of that quotient under completion. This proves the first conclusion over k .

Lemma 4.2 gives the same conclusion over every field of characteristic different from p . Over an arbitrary S , the finite quotient $(\mathrm{MGL}_{S,(p)}[v_n^{-1}]/I_n) \wedge R$ is dualizable over $\mathrm{MGL}_{S,(p)}[v_n^{-1}] \wedge R$ and vanishes at every residue field. By Lemma 4.3, $(\mathrm{MGL}_{S,(p)}[v_n^{-1}]/I_n) \wedge R = 0$ over S . Taking the remaining coefficient cofibers gives $K_S(n) \wedge R \simeq 0$. $\square$

6 Counterexamples with only an $\mathbb{E}_\infty$ structure

Work over $\mathbb{C}$. Choose compatible primitive p -power roots of unity and let

$$\tau : \Sigma^{0,-1} \mathbf{1}_p^\wedge \longrightarrow \mathbf{1}_p^\wedge$$

be the usual motivic Bott class. We write $K(n)/\tau$ for the smash product of $K(n)$ with the cofiber of τ . The scalar p -cofiber in (1.1) ensures that smashing $K(n)$ with $\mathbf{1}_p^\wedge$ leaves it unchanged. Thus there is a specified quotient map $K(n) \rightarrow K(n)/\tau$ and a specified unit on the quotient.

Theorem 6.1. *For every prime p and integer $n \geq 1$, the motivic spectrum $K_{\mathbb{C}}(n)/\tau$ admits an $\mathbb{E}_\infty$ -MGL-algebra structure and satisfies*

$$\begin{aligned} K_{\mathbb{C}}(m) \wedge (K_{\mathbb{C}}(n)/\tau) &\simeq 0 & (0 \leq m < n), \\ K_{\mathbb{C}}(n) \wedge (K_{\mathbb{C}}(n)/\tau) &\not\simeq 0. \end{aligned}$$

Consequently adjacent-height vanishing fails for motivic $\mathbb{E}_\infty$ -rings even over $\mathbb{C}$.

Proof. Bachmann–Hahn construct the $\mathbb{E}_\infty$ -MGL-algebra structure with the specified underlying module and unit [6, Example 1.5]. They also compute

$$\pi_{**}(K(n)/\tau) = \mathbb{F}_p[v_n^{\pm 1}], \quad |v_n| = (2p^n - 2, p^n - 1). \quad (6.1)$$

Thus $K(n)/\tau$ is nonzero. Its rationalization is zero, which proves the assertion for $m = 0$.

For $1 \leq m < n$, the $\mathrm{MGL}_{(p)}$ -module structures on the two factors give an action

$$(\mathrm{MGL}_{(p)} \wedge \mathrm{MGL}_{(p)}) \wedge (K(m) \wedge (K(n)/\tau)) \longrightarrow K(m) \wedge (K(n)/\tau),$$

using the first coefficient factor on $K(m)$ and the second on $K(n)/\tau$. Write v_i^L and v_i^R for the images of v_i under $1 \wedge \eta$ and $\eta \wedge 1$, respectively, where $\eta : \mathbf{1} \rightarrow \mathrm{MGL}_{(p)}$ is the unit. Each image acts as a map from the $(2p^i - 2, p^i - 1)$ suspension of the displayed module to itself. The two orientations of $\mathrm{MGL}_{(p)} \wedge \mathrm{MGL}_{(p)}$ are related by a strict isomorphism of formal group laws [14, Lemma 6.4 and Corollary 6.7]. Invariance of the height ideal gives

$$(p, v_1^L, \dots, v_m^L) = (p, v_1^R, \dots, v_m^R). \quad (6.2)$$

This is an equality of ideals in $\pi_{**}(\mathrm{MGL}_{(p)} \wedge \mathrm{MGL}_{(p)})$. Every generator on the right maps to zero in (6.1). Since $K(n)/\tau$ is an algebra over $\mathrm{MGL}_{(p)}$, these classes act by nullhomotopic module

maps. By (6.2), the left action of v_m is also null. Since v_m acts invertibly on $K(m)$, we obtain $K(m) \wedge (K(n)/\tau) = 0$.

To prove nonvanishing at height n , compose the unit, quotient map, and multiplication:

$$\begin{aligned} K(n)/\tau &\xrightarrow{\text{unit} \wedge 1} K(n) \wedge (K(n)/\tau) \\ &\xrightarrow{\text{quotient} \wedge 1} (K(n)/\tau) \wedge (K(n)/\tau) \xrightarrow{\text{multiplication}} K(n)/\tau. \end{aligned}$$

The quotient map preserves the specified unit, so the composite is the identity. Thus the nonzero spectrum $K(n)/\tau$ is a retract of $K(n) \wedge (K(n)/\tau)$, which is therefore nonzero. $\square$

The two coefficient actions are needed because the theorem concerns the absolute smash product $K(m) \wedge (K(n)/\tau)$. Vanishing of $K(m) \otimes_{\text{MGL}} K(n)$ alone would not prove that assertion.

These examples cannot be normed. Indeed, $\text{HZ} \wedge K(n) \simeq 0$ by [6, Example 1.5], and hence $\text{HZ} \wedge (K(n)/\tau) \simeq 0$. If the quotient were normed, [6, Theorem 1.2] would give $\text{MGL} \wedge (K(n)/\tau) \simeq 0$. The MGL-algebra unit and action exhibit $K(n)/\tau$ as a retract of that product, contradicting (6.1).

7 Counterexamples at the characteristic

We use the coefficient model (1.1) over $\mathbb{F}_p$. In characteristic p , degree-zero scalar quotients and localizations admit normed structures. Applying these constructions to periodic algebraic cobordism will give counterexamples at every positive adjacent pair of heights. We also give a height-zero example at $p = 2$. For the positive-height examples, we need to know that $K_{\mathbf{F}_p}(n)$ is nonzero. We prove this by mapping its zeroth slice to HZ/p .

7.1 Normed quotients and localizations of degree-zero elements

Proposition 7.1. *Let S be an $\mathbf{F}_p$ -scheme and let A be a p -local normed spectrum over S . For every $a \in \pi_{0,0}A$, the specified scalar cofiber A/a and the ordinary commutative localization $A[a^{-1}]$ admit normed structures under A , compatible with their canonical maps from A .*

Proof. Work in $\text{SH}(S)_{(p)}$. Let $U : \text{NAlg}(\text{SH}(S)_{(p)}) \rightarrow \text{SH}(S)_{(p)}$ be the forgetful functor and NSym its left adjoint. The adjunction identifies maps $\text{NSym}(\mathbf{1}) \rightarrow A$ of normed algebras with maps $\mathbf{1} \rightarrow UA$. Its unit defines the generator $x \in \pi_{0,0}U\text{NSym}(\mathbf{1})$. For a representative of a , write $A_{x=a}$ for A equipped with the corresponding evaluation map $\text{NSym}(\mathbf{1}) \rightarrow A$; $\mathbf{1}_{x=0}$ uses the zero evaluation.

The free normed algebra formula [7, Remark 16.27] and [1, Proposition 2.1(3)] identify

$$U\text{NSym}(\mathbf{1}) \simeq \bigoplus_{d \geq 0} (\Sigma_+^\infty B_{\text{ét}} \Sigma_d)_{(p)} \simeq \bigoplus_{d \geq 0} \mathbf{1}.$$

The equivalences of the summands with $\mathbf{1}$ come from their projections to S . Multiplication by x raises the cardinality grading by one and is the identity from each summand to the next.

The zero evaluation gives a nullhomotopy of the composite

$$U\text{NSym}(\mathbf{1}) \xrightarrow{x} U\text{NSym}(\mathbf{1}) \longrightarrow U\mathbf{1}_{x=0}$$

in $\text{Mod}_{U\text{NSym}(\mathbf{1})}(\text{SH}(S)_{(p)})$. It therefore induces a map in that module category

$$\text{cofib}(x : U\text{NSym}(\mathbf{1}) \rightarrow U\text{NSym}(\mathbf{1})) \longrightarrow U\mathbf{1}_{x=0}.$$

The calculation of multiplication by x makes this an equivalence, compatible with the augmentation. Form the pushout of normed algebras

$$A_{x=a} \amalg_{\mathrm{NSym}(\mathbf{1})} \mathbf{1}_{x=0}.$$

Its underlying commutative algebra is the corresponding tensor product [7, Proposition 7.6(3) and Remark 7.9]. Extending scalars along the evaluation $U \mathrm{NSym}(\mathbf{1}) \rightarrow UA$ therefore identifies its underlying module with

$$UA \otimes_{U \mathrm{NSym}(\mathbf{1})} U \mathbf{1}_{x=0} \simeq \mathrm{cofib}(a : UA \rightarrow UA).$$

This equivalence identifies the map from A with the cofiber map and the nullhomotopy of a with the one specified by the quotient.

For localization, apply the same construction to $ax - 1$ in $A \amalg \mathrm{NSym}(\mathbf{1})$. Its underlying module before taking the quotient is $\bigoplus_{d \geq 0} UA$, indexed by powers of x . If $\iota_d : UA \rightarrow \bigoplus_{e \geq 0} UA$ is the d th inclusion, multiplication by $ax - 1$ is the map specified by

$$(ax - 1)\iota_d = \iota_{d+1}a - \iota_d.$$

Its cofiber is the telescope $\mathrm{colim}(UA \xrightarrow{a} UA \xrightarrow{a} \cdots)$. In the quotient algebra, x is an inverse of a , so the map from A factors through $A[a^{-1}]$. The telescope calculation identifies this factorization with an equivalence. $\square$

7.2 The height-zero Moore example

Proposition 7.2. *Over $\mathbb{F}_2$, the Moore spectrum $\mathbf{1}/2$ admits a normed structure and satisfies*

$$\mathrm{H}\mathbb{Q} \wedge \mathbf{1}/2 \simeq 0, \quad (\mathrm{KGL}/2) \wedge \mathbf{1}/2 \not\simeq 0.$$

At $p = 2$, where the usual height-one model is $K(1) \simeq \mathrm{KGL}/2$, this is a counterexample to the height-zero implication.

Proof. Apply Proposition 7.1 to $A = \mathbf{1}_{(2)}$ and $a = 2$. Multiplication by 2 is invertible on $\mathrm{H}\mathbb{Q}$, so $\mathrm{H}\mathbb{Q} \wedge \mathbf{1}/2 \simeq 0$. The equalities $K_0(\mathbb{F}_2) = \mathbb{Z}$ and $K_{-1}(\mathbb{F}_2) = 0$ give

$$\pi_{0,0}(\mathrm{KGL}/2) = \mathbb{F}_2$$

by the cofiber sequence for multiplication by 2 on KGL . The next such cofiber sequence injects $\mathbb{F}_2$ into $\pi_{0,0}((\mathrm{KGL}/2)/2)$, proving nonvanishing. The comparison $K(1) \simeq \mathrm{KGL}/2$ follows from [15, Theorem 5.2 and Remark 5.3]; its finite-dimensional Noetherian base hypothesis holds for $\mathrm{Spec} \mathbb{F}_2$ and does not require 2 to be invertible. $\square$

7.3 Nonvanishing of the Morava quotient

Proposition 7.3. *For the model (1.1), $K_{\mathbb{F}_p}(n) \neq 0$ for every finite $n \geq 1$.*

Frankland–Spitzweck construct retractions onto the expected slices of MGL in characteristic p [10, Theorem 6.26]. To pass from MGL to $K_{\mathbb{F}_p}(n)$, we need these maps to commute with the coefficient cofibers and the v_n telescope. The next two lemmas construct the graded coefficient action and identify the specialization map as a map of modules for that action. The proof of the proposition then computes the resulting map on zeroth slices.

Take the trait $\mathrm{Spec} \mathbf{Z}_p$, with closed and generic points

$$i : \mathrm{Spec} \mathbf{F}_p \longrightarrow \mathrm{Spec} \mathbf{Z}_p, \quad j : \mathrm{Spec} \mathbf{Q}_p \longrightarrow \mathrm{Spec} \mathbf{Z}_p.$$

For each of these bases B , let $\mathrm{SH}(B)^{\mathrm{eff}}$ be the stable localizing subcategory generated by suspension spectra of smooth B -schemes. The effective cover $f_q X \rightarrow X$ is universal among maps to X from $\Sigma^{2q,q} \mathrm{SH}(B)^{\mathrm{eff}}$. Equivalently, it is the counit for the inclusion of this subcategory and its right adjoint. Set

$$f^q X = \mathrm{cofib}(f_{q+1} X \rightarrow X), \quad s_q X = \mathrm{cofib}(f_{q+1} X \rightarrow f_q X).$$

Work in the category of integer-graded motivic spectra $\mathrm{Fun}(\mathbb{Z}_{\mathrm{disc}}, \mathrm{SH}(B))$, with tensor product

$$(M \otimes N)_q = \bigoplus_{r+s=q} M_r \wedge N_s.$$

The full subcategory of objects supported in nonnegative degrees is closed under this tensor product. Let $\mathrm{Sym}_B^{\mathrm{gr}}(t_1, t_2, \dots)$ be the free commutative algebra on the graded object whose degree- d component is $\Sigma^{2d,d} \mathbf{1}_B$ for $d \geq 1$ and whose other components are zero. Its modules are taken in the integer-graded category. Choose representatives of the Lazard classes $t_d : \Sigma^{2d,d} \mathbf{1}_B \rightarrow \mathrm{MGL}_{B,(p)}$ over the trait, and use their pullbacks over its two points. Write $\mathrm{Loc}(\mathbf{1}_B)$ for the full stable localizing subcategory of $\mathrm{SH}(B)$ generated by the unit.

Lemma 7.4 (The graded coefficient action). *The chosen Lazard classes define a map of graded commutative algebras*

$$\mathrm{Sym}_B^{\mathrm{gr}}(t_1, t_2, \dots) \longrightarrow (f_d \mathrm{MGL}_{B,(p)})_{d \geq 0}$$

for each of the three bases B . Moreover,

$$\mathrm{Sym}_B^{\mathrm{gr}}(t_1, t_2, \dots)_d \in \Sigma^{2d,d} \mathrm{Loc}(\mathbf{1}_B). \quad (7.1)$$

Over $\mathbf{Q}_p$, the truncation maps

$$(\mathrm{MGL}_{\mathbf{Q}_p,(p)})_{q \in \mathbb{Z}} \longrightarrow (f^q \mathrm{MGL}_{\mathbf{Q}_p,(p)})_{q \in \mathbb{Z}}$$

form a map of graded modules over this free algebra. The source has one copy of $\mathrm{MGL}_{\mathbf{Q}_p,(p)}$ in every internal degree; its degree- (d, q) action is the composite of the coefficient map to $f_d \mathrm{MGL}_{\mathbf{Q}_p,(p)}$ with multiplication

$$f_d \mathrm{MGL}_{\mathbf{Q}_p,(p)} \wedge \mathrm{MGL}_{\mathbf{Q}_p,(p)} \longrightarrow \mathrm{MGL}_{\mathbf{Q}_p,(p)}.$$

The action on the target is induced by

$$f_d \mathrm{MGL}_{\mathbf{Q}_p,(p)} \wedge f^q \mathrm{MGL}_{\mathbf{Q}_p,(p)} \longrightarrow f^{q+d} \mathrm{MGL}_{\mathbf{Q}_p,(p)}.$$

Proof. The representative of t_d has a d -effective source, so it lifts through $f_d \mathrm{MGL}_{B,(p)}$ with a contractible space of choices. Multiplication on $\mathrm{MGL}_{B,(p)}$ restricts, by the effective-cover adjunction, to maps

$$f_d \mathrm{MGL}_{B,(p)} \wedge f_e \mathrm{MGL}_{B,(p)} \longrightarrow f_{d+e} \mathrm{MGL}_{B,(p)}.$$

The same adjunction lifts the unit, multiplication identities, and their coherences. This gives the displayed family of effective covers its graded commutative algebra structure, to which the chosen generators extend by the free-algebra adjunction.

For a finite-support sequence $\mathbf{m} = (m_1, m_2, \dots)$ of nonnegative integers, put $G_{\mathbf{m}} = \prod_{e \geq 1} \Sigma_{m_e}$. The degree- d component of the free algebra is

$$\mathrm{Sym}_B^{\mathrm{gr}}(t_1, t_2, \dots)_d \simeq \bigoplus_{\sum_e e m_e = d} \left(\bigwedge_{e \geq 1} (\Sigma^{2e, e} \mathbf{1}_B)^{\wedge m_e} \right)_{hG_{\mathbf{m}}}.$$

Here $BG_{\mathbf{m}} \rightarrow \mathrm{SH}(B)$ sends its sole object to the displayed tensor product, and Σ_{m_e} permutes its m_e factors of $\Sigma^{2e, e} \mathbf{1}_B$ through the symmetric monoidal structure. The notation $(-)_{hG_{\mathbf{m}}}$ means the colimit of this diagram. After desuspension by $\Sigma^{2d, d}$, its object is equivalent to $\mathbf{1}_B$, with the induced permutation action retained. Since $\mathrm{Loc}(\mathbf{1}_B)$ is full and closed under colimits, the desuspended homotopy orbits belong to it. This proves (7.1), including the permutation action at $p = 2$.

For $d \geq 0$ and $q \in \mathbb{Z}$, multiplication and the cover maps give a commutative square

$$\begin{array}{ccc} f_d \mathrm{MGL}_{\mathbf{Q}_p, (p)} \wedge f_{q+1} \mathrm{MGL}_{\mathbf{Q}_p, (p)} & \longrightarrow & f_d \mathrm{MGL}_{\mathbf{Q}_p, (p)} \wedge \mathrm{MGL}_{\mathbf{Q}_p, (p)} \\ \downarrow & & \downarrow \\ f_{q+d+1} \mathrm{MGL}_{\mathbf{Q}_p, (p)} & \longrightarrow & \mathrm{MGL}_{\mathbf{Q}_p, (p)}. \end{array}$$

The left vertical map is the lift of multiplication from its $(q + d + 1)$ -effective source. Taking horizontal cofibers gives

$$f_d \mathrm{MGL}_{\mathbf{Q}_p, (p)} \wedge f^q \mathrm{MGL}_{\mathbf{Q}_p, (p)} \longrightarrow f^{q+d} \mathrm{MGL}_{\mathbf{Q}_p, (p)}.$$

The space of lifts in the square is contractible by the effective-cover adjunction. For several factors, the source of the corresponding multiplication map is effective in the sum of their degrees. Applying the same adjunction to those maps lifts the associativity and unit identities and all their higher homotopies. Thus the maps of arrows form a graded module action; their cofibers inherit that action, and the quotient maps from the constant family of copies of MGL are linear. $\square$

Let j_* be the right adjoint of j^* . The functor

$$i^* j_* : \mathrm{SH}(\mathbf{Q}_p) \longrightarrow \mathrm{SH}(\mathbf{F}_p)$$

is lax symmetric monoidal. Apply the adjunction unit $X \rightarrow j_* j^* X$ to the free graded algebra over $\mathbf{Z}_p$, then pull back by i . Since the chosen generators pull back to those at the two points, this gives the map of graded algebras

$$\begin{aligned} \mathrm{Sym}_{\mathbf{F}_p}^{\mathrm{gr}}(t_1, t_2, \dots) &\simeq i^* \mathrm{Sym}_{\mathbf{Z}_p}^{\mathrm{gr}}(t_1, t_2, \dots) \\ &\longrightarrow i^* j_* \mathrm{Sym}_{\mathbf{Q}_p}^{\mathrm{gr}}(t_1, t_2, \dots). \end{aligned}$$

Here $i^* j_*$ is applied in each internal degree. Its lax monoidal structure sends a graded coefficient module over $\mathbf{Q}_p$ to a module over the algebra on the right; restriction along this map makes it a module over the algebra on the left.

Lemma 7.5 (Specialization on slices). *The adjunction unit and the coeffective truncations induce a map of graded modules over $\mathrm{Sym}_{\mathbf{F}_p}^{\mathrm{gr}}(t_1, t_2, \dots)$:*

$$(s_q \mathrm{MGL}_{\mathbf{F}_p, (p)})_q \longrightarrow (s_q i^* j_* f^q \mathrm{MGL}_{\mathbf{Q}_p, (p)})_q. \quad (7.2)$$

The degree- q component of the target is

$$s_q i^* j_* f^q \mathrm{MGL}_{\mathbf{Q}_p, (p)} \simeq \begin{cases} \Sigma^{2q, q} \mathrm{HZ}_{(p)} \otimes_{\mathbf{Z}_{(p)}} L_{(p), q}, & q \geq 0, \\ 0, & q < 0, \end{cases}$$

where $L_{(p), q}$ is the degree- q component of the Lazard ring. These equivalences identify the full graded module action with multiplication in that ring. To view a class $y \in L_{(p), q}$ in the source, lift its representative $\Sigma^{2q, q} \mathbf{1} \rightarrow \mathrm{MGL}_{\mathbf{F}_p, (p)}$ through $f_q \mathrm{MGL}_{\mathbf{F}_p, (p)}$ and compose with the quotient to $s_q \mathrm{MGL}_{\mathbf{F}_p, (p)}$. The displayed map sends this class to y in the corresponding summand of the target coefficients.

Proof. The map before taking slices is the composite

$$\mathrm{MGL}_{\mathbf{F}_p, (p)} \simeq i^* \mathrm{MGL}_{\mathbf{Z}_p, (p)} \longrightarrow i^* j_* \mathrm{MGL}_{\mathbf{Q}_p, (p)} \longrightarrow i^* j_* f^q \mathrm{MGL}_{\mathbf{Q}_p, (p)}.$$

The map to $i^* j_* \mathrm{MGL}_{\mathbf{Q}_p, (p)}$ comes from the adjunction unit over the trait. The final map comes from coeffective truncation. Lax monoidality and Lemma 7.4 make these maps linear over the free coefficient algebra.

We must check that applying s_q in degree q preserves this linearity. For $C \in \mathrm{Loc}(\mathbf{1}_{\mathbf{F}_p})$, the effective-cover adjunction defines

$$C \wedge f_q X \longrightarrow f_q(C \wedge X).$$

The map is an equivalence for $C = \mathbf{1}_{\mathbf{F}_p}$. Over a field, the effective subcategory is generated by objects compact in SH, so f_q preserves coproducts and, being exact, all colimits. Both sides therefore preserve colimits in C , proving the equivalence throughout $\mathrm{Loc}(\mathbf{1}_{\mathbf{F}_p})$. The associativity and unit identities lift uniquely from effective sources. Passing to cofibers and using the Tate-shift formula gives

$$s_{q+d}(\Sigma^{2d, d} C \wedge X) \simeq \Sigma^{2d, d} C \wedge s_q X$$

compatibly with those identities and their higher homotopies. Consequently, for a graded coefficient module $(X_q)_q$, its degree- (d, q) action induces the composite

$$\begin{aligned} \mathrm{Sym}_{\mathbf{F}_p}^{\mathrm{gr}}(t_1, t_2, \dots)_d \wedge s_q X_q \\ \simeq s_{q+d}(\mathrm{Sym}_{\mathbf{F}_p}^{\mathrm{gr}}(t_1, t_2, \dots)_d \wedge X_q) \longrightarrow s_{q+d} X_{q+d}. \end{aligned}$$

The equivalence uses (7.1). Its compatibility with tensor products preserves the module identities and their coherences. Applying this construction to the two graded modules above and to their map gives (7.2) as a map of graded modules.

Frankland–Spitzweck compute the target in each degree [10, Proposition 6.28]. Their proof of the retraction shows that the specialization map sends every Lazard class y to y [10, Theorem 6.26 and its proof]. These integral statements pass to p -local coefficients: the effective covers and slices preserve colimits as above, and j_* preserves coproducts by [10, Lemma 7.4]. Since (7.2) is coefficient-linear, t_d acts on the image of y by sending it to the image of $t_d y$. Thus its action on the target coefficients is polynomial multiplication.

The comparison with polynomial multiplication must identify the action maps and their coherences. For $q, d_1, \dots, d_k \geq 0$, put $D = d_1 + \dots + d_k$. The source of a k -fold action is

$$\left(\bigwedge_{j=1}^k \mathrm{Sym}_{\mathbf{F}_p}^{\mathrm{gr}}(t_1, t_2, \dots)_{d_j} \right) \wedge \left(\Sigma^{2q, q} \mathrm{HZ}_{(p)} \otimes_{\mathbf{Z}_{(p)}} L_{(p), q} \right),$$

and its target is $\Sigma^{2(q+D),q+D}\mathbf{HZ}_{(p)} \otimes_{\mathbf{Z}_{(p)}} L_{(p),q+D}$. For each j , expand the degree- d_j component using the homotopy-orbit formula in Lemma 7.4. After desuspension by $\Sigma^{2(q+D),q+D}$, each resulting source summand is

$$(\mathbf{HZ}_{(p)} \otimes_{\mathbf{Z}_{(p)}} L_{(p),q})_{h(G_{\mathbf{m}_1} \times \dots \times G_{\mathbf{m}_k})}, \quad \sum_e em_{j,e} = d_j.$$

We use the p -local version of [10, Lemma 6.14]. Its proof by coeffective adjunction applies because f^0 preserves filtered colimits and $f^0(\mathbf{1}_{(p)}) \simeq \mathbf{HZ}_{(p)}$. In the full category $\mathrm{SH}(\mathbf{F}_p)$ it gives

$$[\mathbf{HZ}_{(p)}, \Sigma^{s,0}\mathbf{HZ}_{(p)}] \cong [\mathbf{1}_{(p)}, \Sigma^{s,0}\mathbf{HZ}_{(p)}] \cong \begin{cases} \mathbb{Z}_{(p)}, & s = 0, \\ 0, & s \neq 0. \end{cases}$$

The last calculation is weight-zero motivic cohomology. Its localization uses compactness of $\mathbf{1}$, since maps from $\mathbf{1}_{(p)}$ to a p -local target agree with maps from $\mathbf{1}$. Thus mapping spaces between finite sums of unshifted $\mathbf{HZ}_{(p)}$ are discrete, with components the corresponding matrices over $\mathbb{Z}_{(p)}$.

Each factor permutation comes from a block permutation of the trivial vector bundle. The $\mathbf{HZ}_{(p)}$ orientation makes its Thom class invariant under bundle isomorphisms, so each such permutation acts by the identity. The discreteness just proved then identifies the full action with the trivial action. Mapping out of the displayed homotopy orbits takes a limit of these discrete spaces; it remains discrete, and its components are determined by the maps on ordinary H_0 of the group-chain complexes. This applies to every k , including the sources for the unit and associativity identities.

For $k = 1$, the coefficient calculation above identifies these matrices with multiplication by the corresponding monomials in $L_{(p)}$. For arbitrary k , their composites are the iterated multiplication matrices. These matrices specify the polynomial module action: on each homotopy-orbit summand of the fixed free coefficient algebra, the action is multiplication by its monomial in $L_{(p)}$. They satisfy the module identities, and each nonempty component of the operation spaces is contractible. The identifications therefore compare the unit, associativity, and all higher coherences of the constructed action with this polynomial action. The discreteness assertion concerns mapping spaces; the mapping spectra can still have negative homotopy groups given by group cohomology. $\square$

Proof of Proposition 7.3. Apply the coefficient cofibers and the v_n telescope of (1.1) to the map of graded modules in Lemma 7.5. Exactness, preservation of filtered colimits, and the Tate-shift formula identify the source with $(s_q K_{\mathbf{F}_p}(n))_q$. For example, the t_d -cofiber in degree q is

$$\mathrm{cofib}(\Sigma^{2d,d} s_{q-d} \mathrm{MGL}_{\mathbf{F}_p,(p)} \xrightarrow{t_d} s_q \mathrm{MGL}_{\mathbf{F}_p,(p)}) \simeq s_q(\mathrm{MGL}_{\mathbf{F}_p,(p)}/t_d).$$

On the target, the p -cofiber replaces each coefficient copy by $\mathbf{HZ}/p$. Multiplication by t_d embeds the monomial basis as the subset divisible by t_d , so its cofiber is the sum on the complementary monomials. Taking the cofibers for all $d \neq p^n - 1$ and the filtered colimit leaves the graded object with component $\Sigma^{2r(p^n-1),r(p^n-1)}\mathbf{HZ}/p$ in degree $r(p^n - 1)$ for $r \geq 0$, indexed by the basis vector v_n^r , and zero in all other degrees. Multiplication by v_n carries the copy indexed by v_n^r to the copy indexed by v_n^{r+1} . In internal degree zero, its localization telescope is therefore

$$\mathbf{HZ}/p \xrightarrow{1} \mathbf{HZ}/p \xrightarrow{1} \dots,$$

The r th term is the component in degree $r(p^n - 1)$, desuspended by $\Sigma^{2r(p^n-1),r(p^n-1)}$.

The unit therefore induces

$$s_0(\mathbf{1}_{\mathbf{F}_p, (p)}) = \mathbf{HZ}_{(p)} \longrightarrow s_0 K_{\mathbf{F}_p}(n) \longrightarrow \mathbf{HZ}/p, \quad 1 \longmapsto 1.$$

The composite is nonzero, so $s_0 K_{\mathbf{F}_p}(n)$, and hence $K_{\mathbf{F}_p}(n)$, is nonzero. Only the functor s_0 is used here; the periodic spectrum need not be effective. $\square$

7.4 Failure at every positive adjacent pair

Use the Thom normed structure on PMGL. Its underlying MGL-module is $\bigoplus_{r \in \mathbb{Z}} \Sigma^{2r, r} \text{MGL}$, with the r th summand coming from virtual bundles of rank r . This is a decomposition of MGL-modules: for $c \in \pi_{2d, d} \text{MGL}$, the scalar map on the rank- r summand is

$$\Sigma^{2r, r}(c : \Sigma^{2d, d} \text{MGL} \longrightarrow \text{MGL}).$$

Thus the original coefficient scalars preserve each rank summand.

Theorem 7.6. *For every prime p and every finite $h \geq 1$, the spectrum*

$$\text{PMGL}_{\mathbf{F}_p, (p)}[v_{h+1}^{-1}]/(p, v_1, \dots, v_h)$$

admits a normed structure and satisfies

$$\begin{aligned} K_{\mathbf{F}_p}(h) \wedge \text{PMGL}_{\mathbf{F}_p, (p)}[v_{h+1}^{-1}]/(p, v_1, \dots, v_h) &= 0, \\ K_{\mathbf{F}_p}(h+1) \wedge \text{PMGL}_{\mathbf{F}_p, (p)}[v_{h+1}^{-1}]/(p, v_1, \dots, v_h) &\neq 0. \end{aligned}$$

Both products are absolute smash products in $\text{SH}(\mathbf{F}_p)$.

Proof. The Thom normed structure on PMGL is constructed in [7, Theorem 16.19]. If $u \in \pi_{-2, -1} \text{PMGL}$ is the inverse period, then $v_j u^{p^j - 1}$ has degree zero. Proposition 7.1 allows us to invert $v_{h+1} u^{p^{h+1} - 1}$ and quotient by p and $v_j u^{p^j - 1}$ for $1 \leq j \leq h$, as normed algebras. Multiplication by each invertible power of u identifies the source of the corresponding degree-zero scalar map with the source of $v_j : \Sigma^{2p^j - 2, p^j - 1} \text{PMGL} \rightarrow \text{PMGL}$. These source equivalences identify the cofibers and localization telescopes, and hence the underlying module with $\text{PMGL}_{\mathbf{F}_p, (p)}[v_{h+1}^{-1}]/(p, v_1, \dots, v_h)$. In the constructed commutative algebra, p and $v_j u^{p^j - 1}$ for $1 \leq j \leq h$ have specified nullhomotopies. Multiplying by the inverse powers of u makes the images of $p, v_1, \dots, v_h$ zero and their scalar module maps nullhomotopic.

For the height- h smash product, use the two coefficient actions of $\text{MGL}_{(p)} \wedge \text{MGL}_{(p)}$ from (6.2). The two orientations are strictly isomorphic, so

$$(p, v_1^L, \dots, v_h^L) = (p, v_1^R, \dots, v_h^R).$$

This identity follows from the Lazard Hopf algebroid [14, Lemma 6.4 and Corollary 6.7]; a full calculation of motivic cooperations is unnecessary. The right-hand ideal acts trivially on the quotient of PMGL, so v_h^L acts trivially on the smash product. Its action through $K_{\mathbf{F}_p}(h)$ is invertible. The height- h product is therefore zero.

For height $h+1$, first compute the relative tensor product. The rank decomposition commutes with the cofibers of the original v_j , since their scalar maps preserve each summand. Relative

tensoring with $K_{\mathbf{F}_p}(h+1)$ then makes the v_{h+1} -localization map an equivalence, because that scalar already acts invertibly on $K_{\mathbf{F}_p}(h+1)$. We obtain

$$\begin{aligned} K_{\mathbf{F}_p}(h+1) \otimes_{\mathrm{MGL}_{\mathbf{F}_p, (p)}} (\mathrm{PMGL}_{\mathbf{F}_p, (p)}[v_{h+1}^{-1}]/(p, v_1, \dots, v_h)) \\ \simeq \bigoplus_{r \in \mathbf{Z}} \Sigma^{2r, r} (K_{\mathbf{F}_p}(h+1)/(p, v_1, \dots, v_h)). \end{aligned}$$

Each of $p, v_1, \dots, v_h$ acts nilpotently on $K_{\mathbf{F}_p}(h+1)$. Indeed, a scalar acts square-zero on its own cofiber, and the subsequent coefficient cofibers and telescope preserve that relation as module functors. The relation also persists after the additional finite quotients in the displayed formula.

A nilpotent endomorphism of a nonzero object has nonzero cofiber. Otherwise it would be an equivalence, and a zero power would also be an equivalence. Starting with Proposition 7.3 and applying this observation to the successive scalar quotients proves that the relative tensor product is nonzero.

If the absolute smash product at height $h+1$ were zero, the two-sided simplicial bar computing the relative tensor product would be degreewise zero. For $s \geq 0$, its degree- s term is

$$K_{\mathbf{F}_p}(h+1) \wedge \mathrm{MGL}_{\mathbf{F}_p, (p)}^{\wedge s} \wedge (\mathrm{PMGL}_{\mathbf{F}_p, (p)}[v_{h+1}^{-1}]/(p, v_1, \dots, v_h)).$$

The face maps use the two module actions and multiplication in $\mathrm{MGL}_{\mathbf{F}_p, (p)}$; the degeneracies insert its unit. Each term is the assumed zero absolute product smashed with additional MGL factors. Its geometric realization would therefore be zero, contradicting the relative calculation. The absolute height- $(h+1)$ product is nonzero. $\square$

The results concern the coefficient models fixed in Section 1. They do not classify counterexamples over each mixed-characteristic base on which p fails to be invertible, or address the spectra $K(\beta_{ij})$ in the second part of Bachmann–Hahn’s question.

A Indexed coefficient colimits

Let E be a normed algebra over a field k , and let U range over smooth k -schemes. Lemma 2.2 uses a Thom–norm comparison that is natural in maps of coefficient diagrams, including maps to zero. We construct that comparison here and identify its cyclic product presentation. The construction uses finite étale distributivity and sifted extension as in [7, §16.3, construction (16.15)].

Let $\mathbf{FEt}_U$ be the category of finite étale U -schemes. The coefficient categories and pullback functors form a functor

$$\mathbf{FEt}_U^{\mathrm{op}} \longrightarrow \mathbf{Cat}_\infty, \quad W \longmapsto \mathrm{Mod}_{E_W}(\mathrm{SH}(W)).$$

Its cartesian Grothendieck construction has objects $(q : W \rightarrow U, M)$, with $M \in \mathrm{Mod}_{E_W}(\mathrm{SH}(W))$. An arrow to $(q' : W' \rightarrow U, M')$ consists of $t : W \rightarrow W'$ over U and an E_W -linear map $M \rightarrow t^*M'$. For composable arrows over t and t' , the coefficient map of their composite is

$$M \longrightarrow t^*M' \longrightarrow t^*(t')^*M'' \simeq (t't)^*M''.$$

A coefficient diagram consists of a small infinity-category D and a functor

$$\ell : D \longrightarrow \int_{W \in \mathbf{FEt}_U} \mathrm{Mod}_{E_W}(\mathrm{SH}(W)), \quad d \longmapsto (q_d : W_d \rightarrow U, \ell_d),$$

with the displayed notation for its values. Define its coefficient colimit by

$$\mathrm{Th}_{E,U}(D, \ell) := \mathrm{colim}_{d \in D} (q_d)_\#^E \ell_d \quad \text{in } \mathrm{Mod}_{E_U}(\mathrm{SH}(U)),$$

where $(q_d)_\#^E$ is left adjoint to pullback on coefficient modules. The maps in this colimit are induced by adjunction from the coefficient maps $\ell_d \rightarrow t^*\ell_{d'}$.

Write $\mathrm{Fam}_E(U)$ for this category of finite étale families of E -modules. We use its sifted completion $\mathcal{P}_\Sigma(\mathrm{Fam}_E(U))$. Here $\mathcal{P}_\Sigma$ consists of the space-valued presheaves that take finite coproducts to products; we use small presheaves if the coefficient category is large. Finite coproducts in $\mathrm{Fam}_E(U)$ are disjoint unions of the parameter schemes, with the prescribed module on each component. Write y_U for the Yoneda functor into $\mathcal{P}_\Sigma(\mathrm{Fam}_E(U))$; it preserves these finite coproducts. Associate to (D, ℓ) the coefficient presheaf

$$P_{D,\ell} = \mathrm{colim}_{d \in D} y_U(q_d, \ell_d) \quad \text{in } \mathcal{P}_\Sigma(\mathrm{Fam}_E(U)).$$

The coefficient-colimit functor extends to a functor $\mathrm{Th}_{E,U} : \mathcal{P}_\Sigma(\mathrm{Fam}_E(U)) \rightarrow \mathrm{Mod}_{E_U}(\mathrm{SH}(U))$, characterized by

$$\mathrm{Th}_{E,U} y_U(q, M) = q_\#^E M.$$

It is left adjoint to $T \mapsto y_U(\mathrm{id}_U, T)$: on a representable, the required equivalence of mapping spaces is

$$\Omega^\infty \mathrm{Map}(q_\#^E M, T) \simeq \Omega^\infty \mathrm{Map}(M, q^*T).$$

Consequently it preserves colimits, with $\mathrm{Th}_{E,U}(P_{D,\ell}) = \mathrm{Th}_{E,U}(D, \ell)$. We use the same symbol $\mathrm{Th}_{E,U}$ for this extension. This is the coefficient version of the construction in [7, Definition 16.1 and §16.3]; the norm-compatible sifted completion is construction (16.15) there.

Here is an explicit presentation of $P_{D,\ell}$. Choose a simplicial-set model of D , and let D_n be its set of n -simplices. Its simplicial replacement has term

$$B_n(D, \ell) = \coprod_{\sigma \in D_n} y_U(q_{\sigma(0)}, \ell_{\sigma(0)}), \quad |B_\bullet(D, \ell)| \simeq P_{D,\ell}. \quad (\mathrm{A.1})$$

The realization formula follows from the finality of the first-vertex map $\Delta_{/D}^{\text{op}} \rightarrow D$ [17, Proposition 12.2 and Corollary 12.3]. For a diagram $\ell : K \rightarrow \text{Fam}_E(U)$ indexed by any small simplicial set, define $P_{K,\ell} = \text{colim}_K y_U \ell$; the same formula computes this colimit. An infinite coproduct in a term of (A.1) is the filtered colimit of its finite subcoproducts. Each such finite subcoproduct is represented by a finite étale parameter scheme.

For a finite étale map $f : V \rightarrow U$, the norm on these coefficient presheaves is denoted $\mathcal{N}_f : \mathcal{P}_\Sigma(\text{Fam}_E(V)) \rightarrow \mathcal{P}_\Sigma(\text{Fam}_E(U))$. On a generating pair $(a : W \rightarrow V, M)$ it is given as follows. Put $Z = R_f W$, with structure map $r : Z \rightarrow U$, projection $f' : Z \times_U V \rightarrow Z$, and evaluation $e : Z \times_U V \rightarrow W$. Then

$$\mathcal{N}_f y_V(a, M) = y_U(r, N_f^E e^* M). \quad (\text{A.2})$$

Weil restriction and the coefficient norm define this functor on the full Grothendieck categories, and the universal sifted completion extends it to the presheaf categories. In particular, $\mathcal{N}_f$ preserves filtered colimits and geometric realizations. Equations (A.1) and (A.2) therefore specify $\mathcal{N}_f P_{D,\ell}$ on arbitrary coefficient diagrams, including all their morphisms.

Lemma A.1 (Indexed coefficient colimits). *The functors $\text{Th}_{E,U}$ commute naturally with pullback and the finite étale norms just defined. In particular,*

$$\begin{aligned} N_f^E \text{Th}_{E,V}(D, \ell) &\simeq \text{Th}_{E,U}(\mathcal{N}_f P_{D,\ell}) \\ &=: \text{Th}_{E,U} \left(\prod_{v \in V/U} D_v, \bigotimes_{v \in V/U} \ell_v \right). \end{aligned} \quad (\text{A.3})$$

The product notation on the second line denotes the coefficient presheaf constructed above; it does not presume a global ordering of the sheets of f . For a split map it is represented by the ordinary product of the diagrams on its sheets, with tensor-product coefficients. The equivalence is natural in morphisms of coefficient presheaves, hence in diagram morphisms, augmentations, and base change. In particular it applies to noninvertible coefficient maps and zero cone vertices.

Proof. On generating pairs, pullback compatibility is the base-change equivalence for $q_\#^E$. For norm compatibility, use the exponential diagram in (A.2). Distributivity [7, Proposition 5.10], the projection formula, and norm base change give the following natural equivalence on the full module category:

$$\begin{aligned} N_f^E a_\#^E M &\simeq E_U \otimes_{f_\otimes E_V} r_\#(f'_\otimes e^* M) \\ &\simeq r_\#(E_Z \otimes_{f'_\otimes E_Z \times_U V} f'_\otimes e^* M) \simeq r_\#^E N_{f'}^E e^* M. \end{aligned}$$

These are the exchange transformations required by the relative-adjunction construction of [7, §16.3, construction (16.15)], now applied to the normed module categories. The finite étale parameter maps are closed under composition, base change, and finite étale Weil restriction. Base change and the displayed distributivity equivalence show that the resulting map of cartesian fibrations preserves cartesian edges. Thus the exchange maps form a natural transformation on the span category, with its composition and base-change coherences. Since both coefficient norms and the functors $\mathcal{N}_f$ preserve sifted colimits, this transformation extends to the presheaf categories and gives a natural equivalence of functors

$$\alpha_f : N_f^E \text{Th}_{E,V} \xrightarrow{\simeq} \text{Th}_{E,U} \mathcal{N}_f.$$

For an arrow $P \rightarrow P'$ in $\mathcal{P}_\Sigma(\text{Fam}_E(V))$, the naturality square of α_f is an equivalence of the specified arrows. This explains compatibility with quotient diagrams: the square retains the actual map $P \rightarrow P'$, including a component $E \rightarrow 0$.

The cyclic orbit presentation identifies this construction with the product diagrams used in Lemma 2.2. Suppose f is a $G = C_p$ -torsor and a finite coproduct has labels in a finite set I , with coefficient M_i over V for $i \in I$. For $W = \coprod_{i \in I} V$ one has

$$R_f W \simeq V \times^G I^p \simeq \coprod_{[\tau] \in I^p/G} V/H_\tau, \quad H_\tau = \text{Stab}_G(\tau).$$

Here G permutes the p coordinates of I^p . Let M be the module on W whose restriction to its i -th component is M_i . The coefficient on each orbit scheme is the restriction of $N_{f'}^E e^* M$ from the exponential diagram. This defines it over V/H_τ directly. On pullback to V , it is the tensor product of the appropriate deck transforms of the M_i . All faces, degeneracies, and coefficient maps are obtained by applying the same functor, so their coherences are retained.

For a finite simplicial model X and a line diagram ℓ on its realization, use $I = X_n$ in degree n of (A.1), where X_n is the set of n -simplices. The resulting degree is indexed by $(X_n)^p = (X^p)_n$, with the permutation action on the coordinates. Their associated finite étale schemes, with the preceding coefficients, give the transported product diagram. Its underlying space parameter is the orbit-associated presheaf of the permutation product: for each subgroup, fixed points commute with realization of these simplicial sets, and the orbit-associated construction preserves colimits. The construction also applies to a categorical zero-cone diagram, but in that case it remains a coefficient presheaf, since its coefficient maps need not be equivalences.

The usual diagonal comparison is valid before orbit transport. For $\delta : \Delta^{\text{op}} \rightarrow (\Delta^{\text{op}})^G$, an object d of the target and $H \leq G_d$, an H -fixed object of $(d \downarrow \delta)$ is a tuple of maps that is constant on each H -orbit of coordinates. Consequently

$$(d \downarrow \delta)^H \cong (\bar{d} \downarrow \delta_{G/H}).$$

The right-hand nerve is contractible because finite diagonals of the sifted category Δ^{op} are final. This verifies equivariant finality, rather than merely ordinary finality. It identifies diagonal and iterated realizations in the orbit presentation, with their coefficients and augmentations. For a split map, separate colimit preservation of tensor product gives the ordinary product diagram asserted in the statement. $\square$

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